Floatway Learning Center



Politeknik Negeri Bandung - **Day One** 26 November 2019

Training Material

4G LTE RF (Introduction to 5G)

FLOATWAY SYSTEMS

Our Product and Service





Learning Center

Research and Development

Industrial Product

Pre Test

1. Please Fill blank column in table below!

Generation	Operating Frequency	Frequency Bandwidth	Multiple Access	Modulation	Technology	Maximum Throughput
1G					AMPS, NMT, TACS	
2G	900 MHz, 1800 MHz	200 kHz				
2,5G				GMSK (GPRS), 8-PSK (EDGE)	GPRS, EDGE	
3G			WCDMA			
3,5G						21 Mbps HSPA+
4G						

2. Explain differences between 2G, 3G and 4G Network Architecture

Floatway Learning Center

FLOATWAY SYSTEMS

Curiculum Vitae

Lingga Wardhana, ST., MBA



Educational Background :

- Electrical Engineering, Gadjah Mada University (2001 – 2006)
- MBA Program, Gadjah Mada University (2011 – 2014)

Professional Experience :

- PT. Siemens Indonesia, Network Optimization Engineer (2006 – 2008)
- PT. Lexcorp Solution, RF Optimization Consultant (2008 – 2009)
- PT. Nexwave, RF Optimization Consultant (2009)
- PT. Lintas Media Telekomunikasi, 3G RF Senior Optimization Consultant (2009 – 2010)
- PT. Floatway Systems, Founder (2010 – Present)

Achievements :

- 1st winner of Indosat Telco Project 2005
- 1st winner of Indosat Wireless Innovation Contest 2007, Hardware Category
- 3rd position in European Satellite Navigation Competititon 2009 for Regional Gipuzkoa/Spain (www.galileo-masters.eu)

Books :

- Belajar Sendiri Mikrokontroler AVR ATMega8535 (2006)
- Technopreneur (2010)
- Teknologi Wireless Communication dan Wireless Broadband (2010)
- 2G/3G RF Planning and Optimization for Consultant (2011)
- 4G Handbook Edisi Bahasa Indonesia (2014)
- Enterpreneurial MBA (2014)

Floatway Learning Center

Participant Introduction

- Name
- Current Job Profile
- Previous Experience
- Expectations, etc.

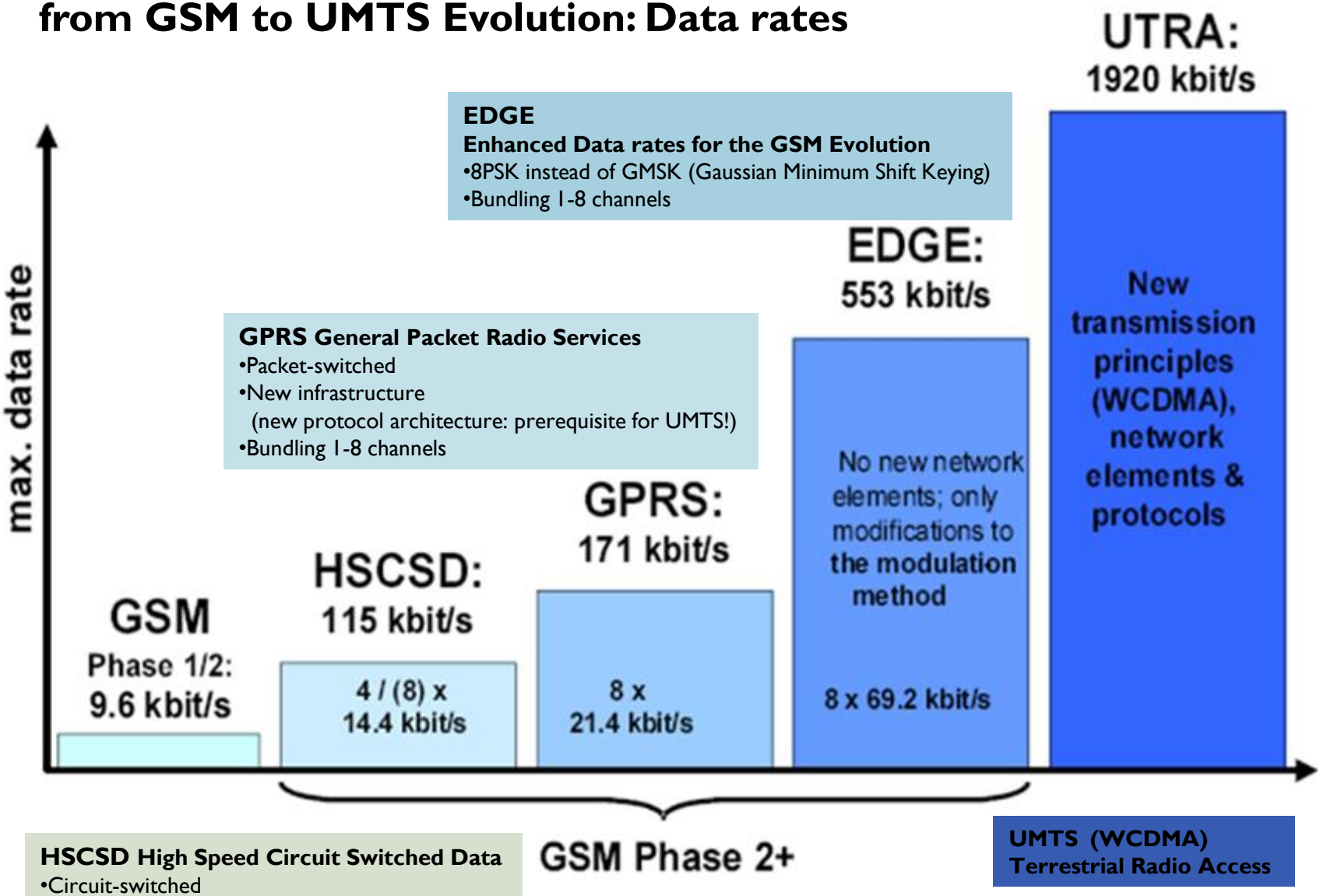


Floatway Learning Center

RADIO CELLULAR TECHNOLOGY

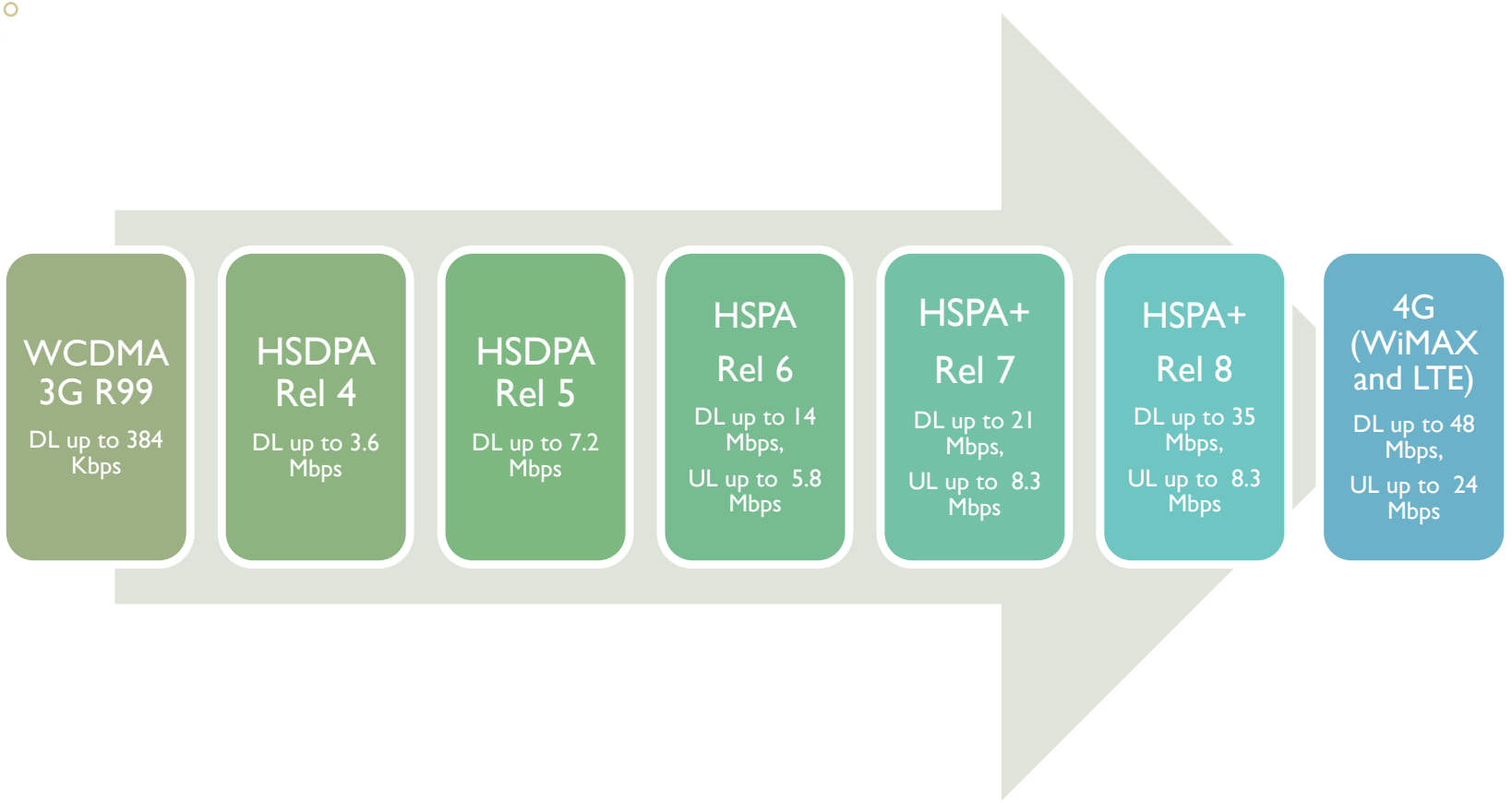
2G & 3G Radio Technology

from GSM to UMTS Evolution: Data rates



Floatway Learning Center

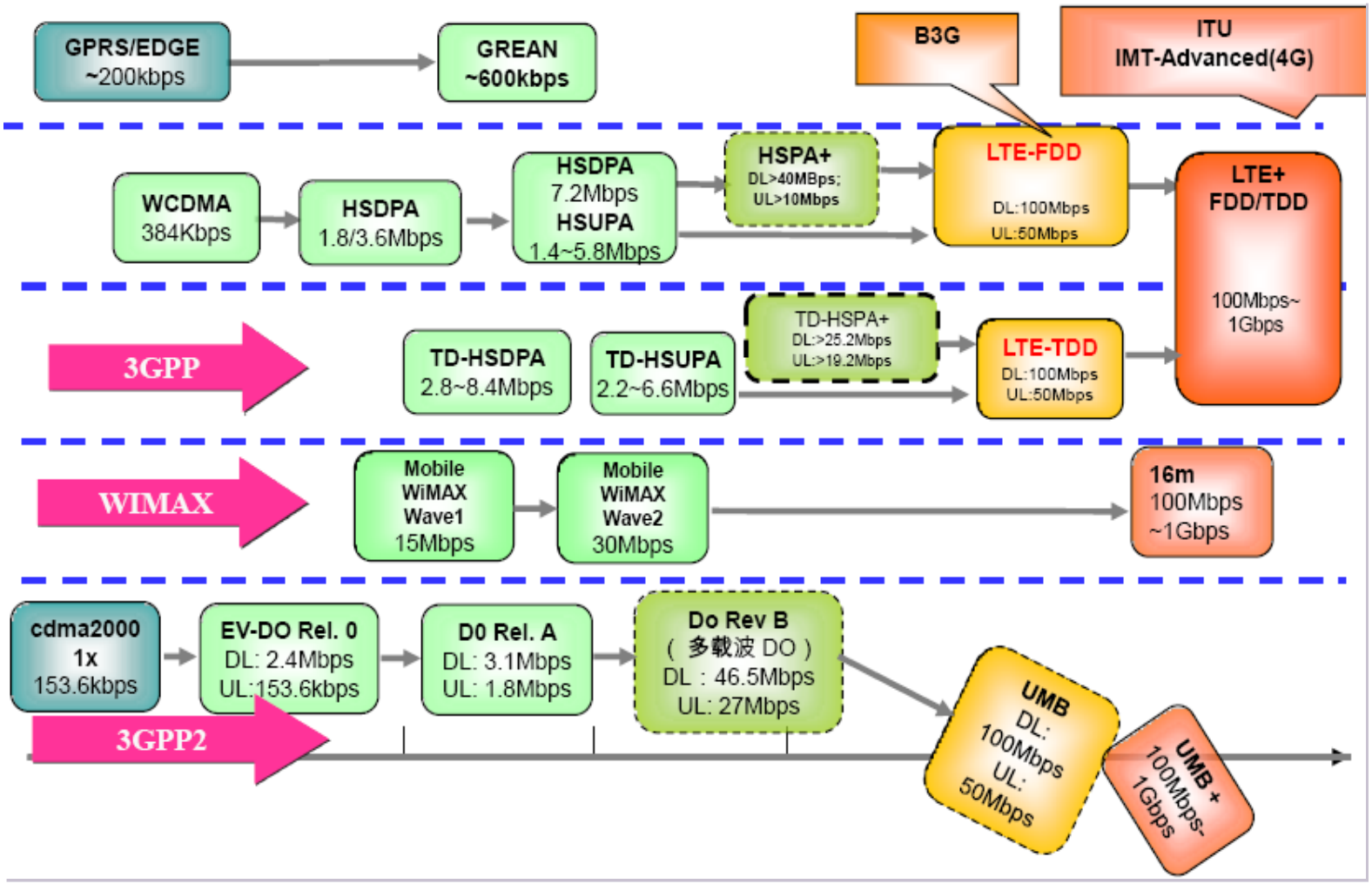
Wireless Broadband Technology Evolution .



Floatway Learning Center

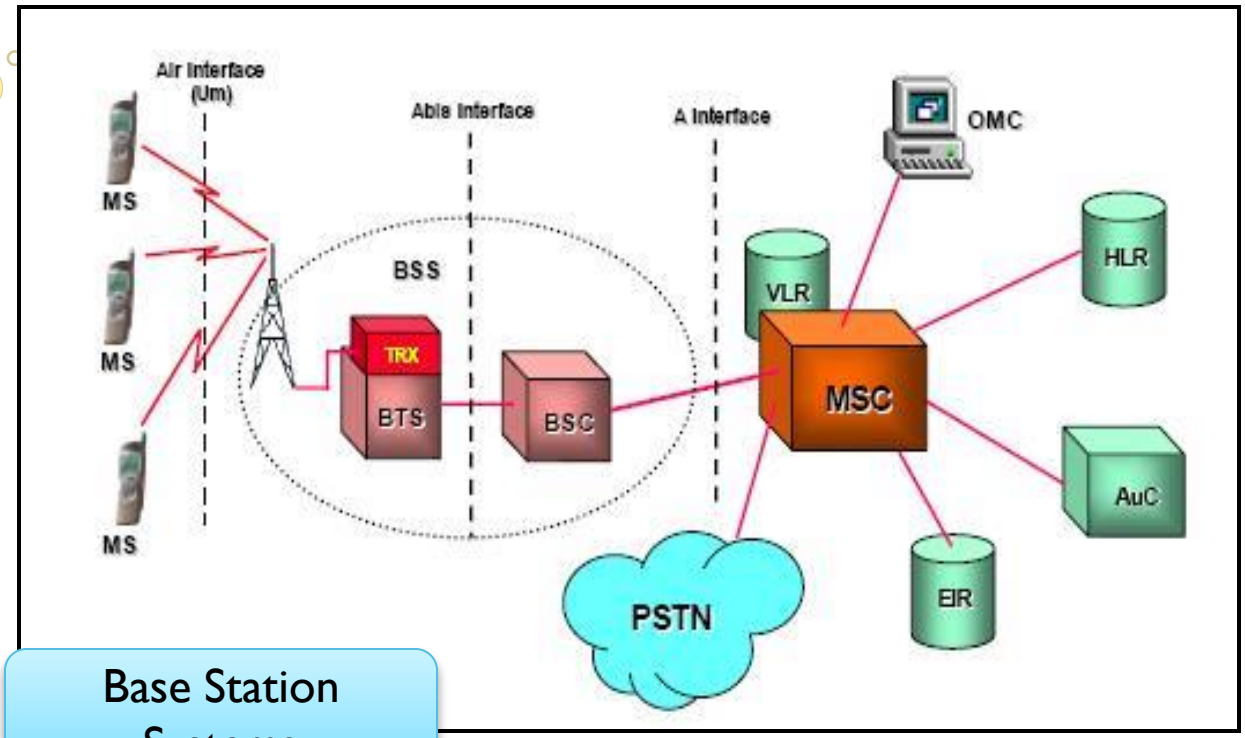
FLOATWAY SYSTEMS

Towards to 4G



NETWORK ARCHITECTURE

GSM Architecture



Network Switching Systems

Mobile Switching Center (MSC)
berfungsi sebagai switch dan penghubung dengan jaringan fixed.

Home Location Register (HLR)
HLR merupakan database yang berisi data-data pelanggan yang tetap.

Visitor Location Register (VLR)
VLR merupakan database yang berisi informasi sementara mengenai pelanggan, terutama saat lokasi dari pelanggan diluar cakupan area jaringan HLR-nya

Base Station Systems

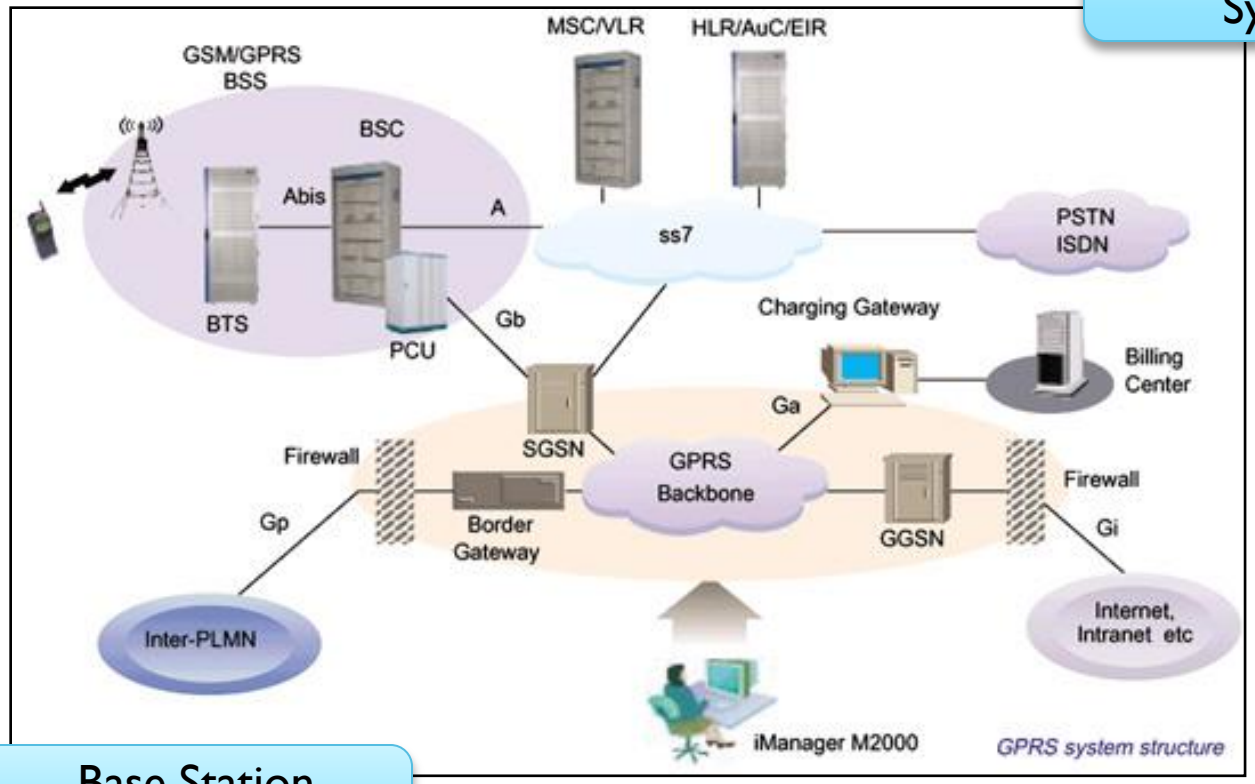
Base Transceiver Station (BTS)
BTS merupakan perangkat pemancar dan penerima yang memberikan pelayanan radio kepada MS.

Base Station Controller (BSC)
BSC membawahi satu atau lebih BTS serta mengatur trafik yang datang dan pergi dari BSC menuju MSC atau BTS.

Floatway Learning Center

FLOATWAY SYSTEMS

GPRS Architecture



Network Switching Systems

GGSN (Gateway GPRS Support Node)
GGSN berfungsi sebagai Gateway antara jaringan GPRS dengan jaringan paket data standar (PDN). lewat antarmuka Gi (Gi Interface) atau ke jaringan GPRS dengan PLMN (Public Land Mobile Network) yang berbeda lewat antarmuka Gp.

Base Station Systems

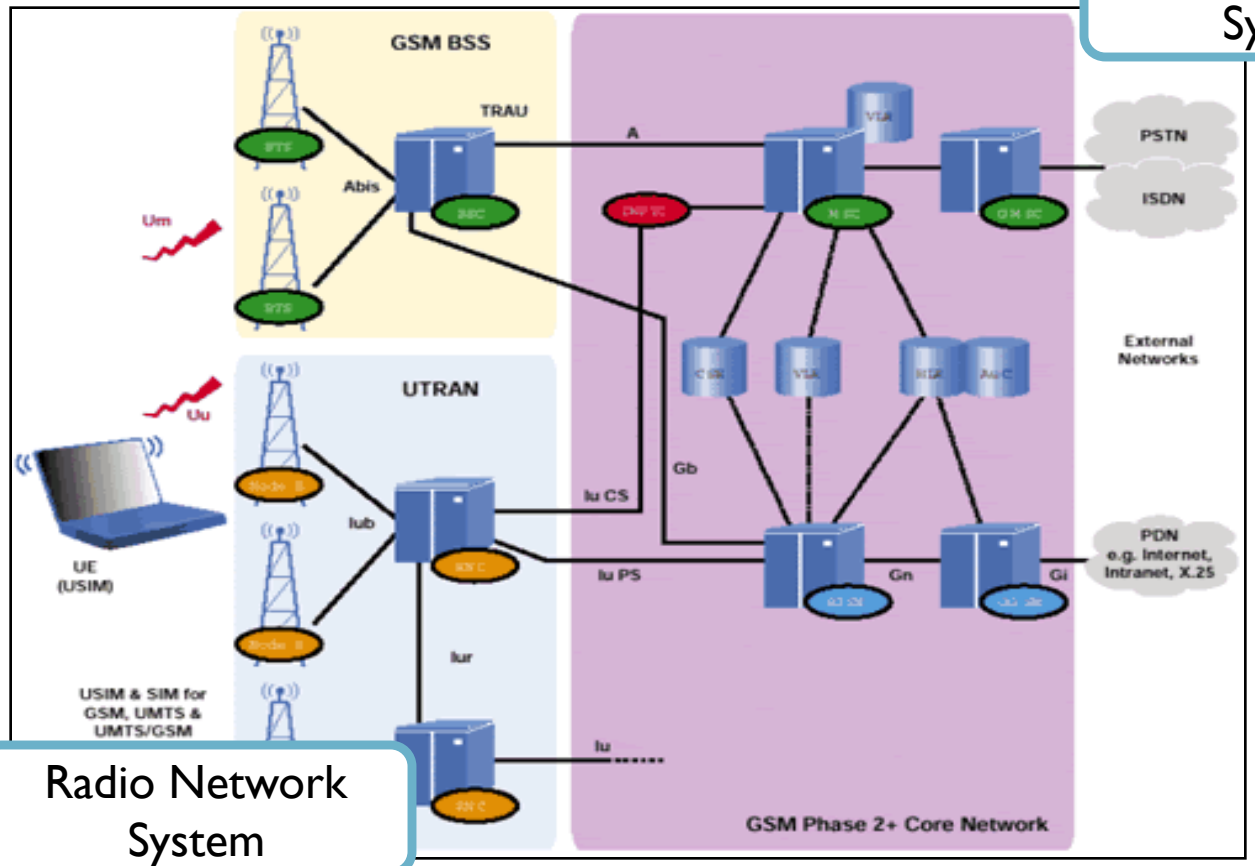
PCU (Packet Control Unit)
Diletakan dalam BSC. bertanggung jawab atas semua protokol radio GPRS dan komunikasi dengan SGSN.

SGSN (Serving GPRS Support Node)
Fungsi SGSN sama seperti fungsi MSC pada jaringan GSM yang berfungsi dalam Mobility Management, Chipperring, kompresi data, paging, perhitungan trafik, charging, security, dan mengatur proses pengaksesan data.

Floatway Learning Center

FLOATWAY SYSTEMS

3G Architecture



Network Switching Systems

Mobile Switching Center (MSC)
berfungsi sebagai switch dan penghubung dengan jaringan fixed.

Home Location Register (HLR)
HLR merupakan database yang berisi data-data pelanggan yang tetap.

Visitor Location Register (VLR)
VLR merupakan database yang berisi informasi sementara mengenai pelanggan, terutama saat lokasi dari pelanggan diluar cakupan area jaringan HLR-nya

Radio Network System

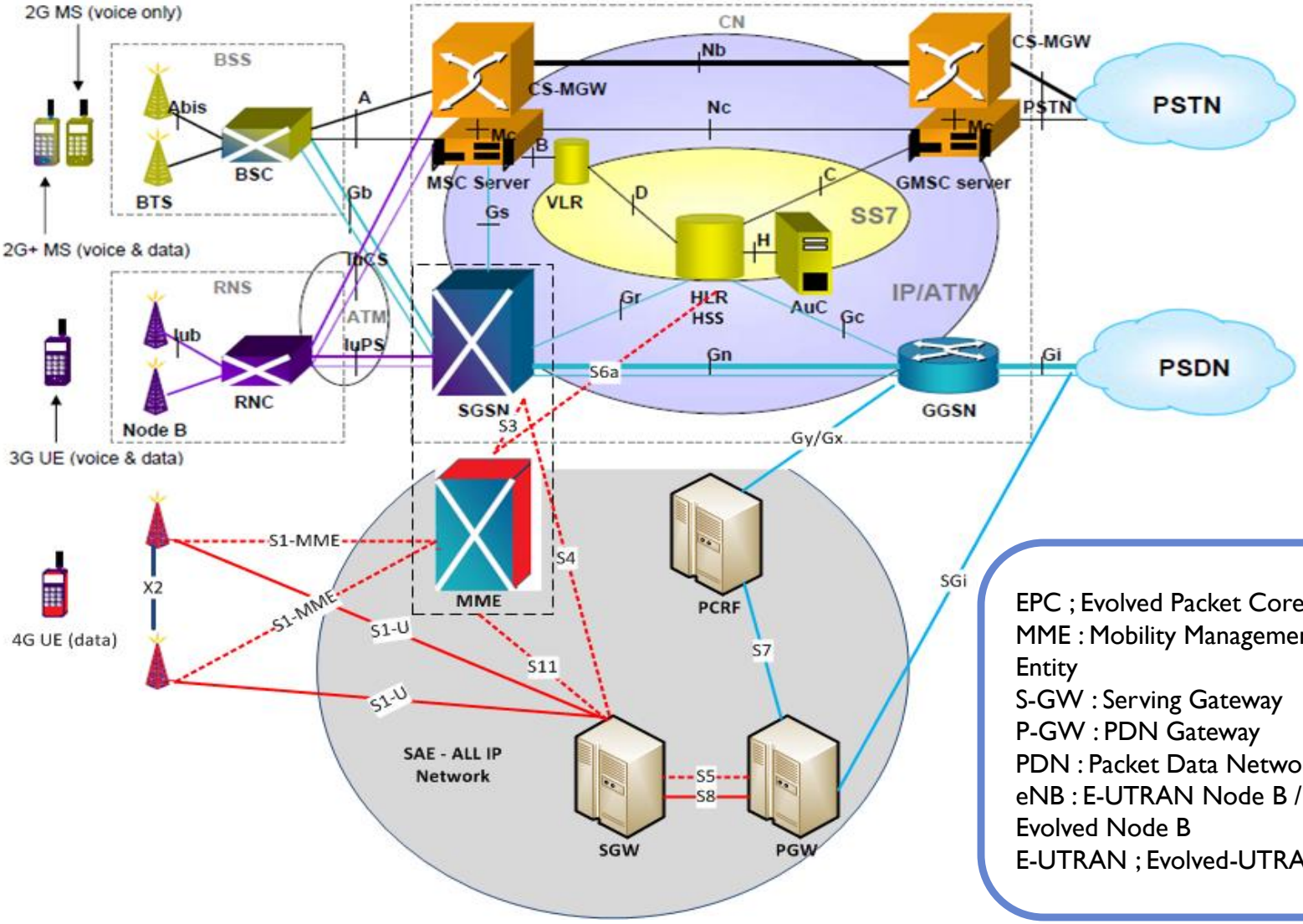
Node B
Node B adalah unit fisik untuk mengirim/menerima frekuensi pada sel.

RNC (Radio Network Controller)
RNC bertanggungjawab manajemen sumber radio yang tersedia pada Node B yang ditanganinya.

Floatway Learning Center

FLOATWAY SYSTEMS

4G Architecture



EPC ; Evolved Packet Core
MME : Mobility Management Entity
S-GW : Serving Gateway
P-GW : PDN Gateway
PDN : Packet Data Network
eNB : E-UTRAN Node B / Evolved Node B
E-UTRAN ; Evolved-UTRAN

Floatway Learning Center

FLOATWAY SYSTEMS

Quiz 1 : Network Architecture

1. Si A (MOC) menelepon si B (MTC). Si A menggunakan jaringan 2G sedangkan si B menggunakan 3G. Gambarkan aliran data A ke B pada arsitektur jaringan seluler !

2. Si A kembali menelepon si B menggunakan Google Talk. Si A menggunakan jaringan 3G sedangkan si B menggunakan 2G. Gambarkan aliran data dari A ke B pada arsitektur jaringan seluler !

3. Si C mengirimkan data kepada si D. Si C menggunakan jaringan 4G sedangkan si D menggunakan 3G. Gambarkan aliran data dari C ke D pada arsitektur jaringan seluler !

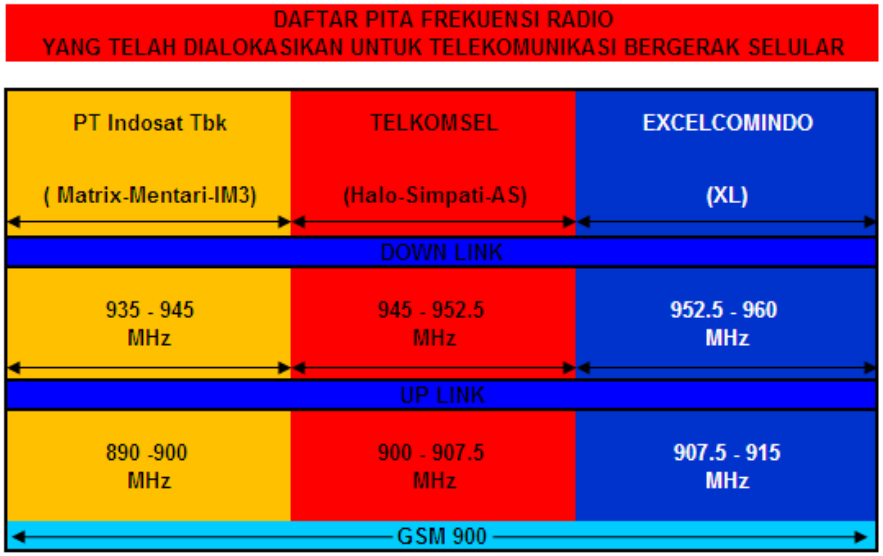
4. Apakah Jaringan 4G LTE dapat menerima panggilan suara (voice call)?

Floatway Learning Center

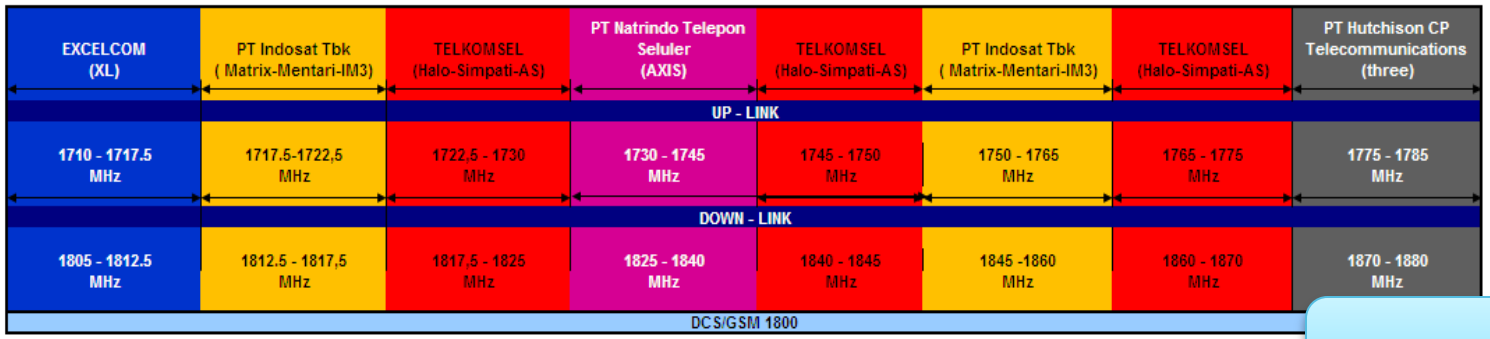
FLOATWAY SYSTEMS

CELLULAR FREQUENCY ALLOCATION

2G Frequency Allocation in Indonesia



GSM 900

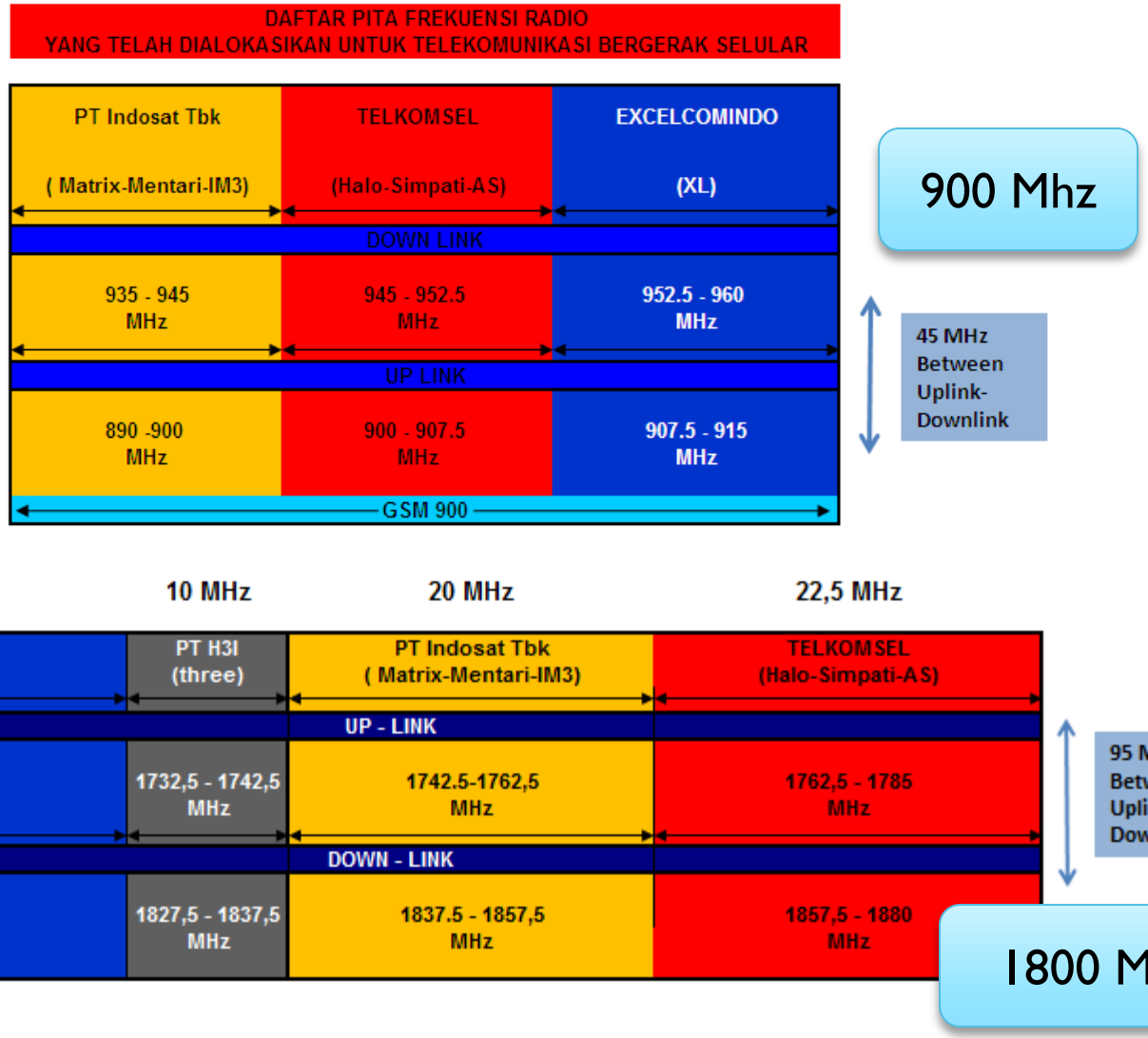


DCS 1800

Floatway Learning Center

FLOATWAY SYSTEMS

900 MHz & 1800 MHz Frequency Allocation



Floatway Learning Center

FLOATWAY SYSTEMS

3G Frequency Allocation in Indonesia

Frequency Spectrum Update March 2013

PITA 3G				
Blok 1	Blok 2	Blok 3	Blok 4	Blok 5
2110 - 2115 Mhz	2115 - 2120 Mhz	2120 - 2125 MHz	2125 - 2130 MHz	2130 - 2135 MHz
PT Hutchison CP Telecommunications (three)	PT Natrindo Telepon Seluler (AXIS)	PT Natrindo Telepon Seluler (AXIS)	TELKOMSEL (Halo-Simpati-AS)	TELKOMSEL (Halo-Simpati-AS)
DOWN LINK				
1920 - 1925 MHz	1925 - 1930 MHz	1930 - 1935 MHz	1935 - 1940 MHz	1940 - 1945 MHz
(first carrier)	(Plan for first carrier)	(second carrier)	(first carrier)	(second carrier)
UPLINK				

PITA 3G						
Blok 6	Blok 7	Blok 8	Blok 9	Blok 10	Blok 11	Blok 12
2135 - 2140 MHz	2140 - 2145 MHz	2145 - 2150 MHz	2150 - 2155 MHz	2155 - 2160 MHz	2160 - 2165 MHz	2165 - 2170 MHz
PT Hutchison CP Telecommunications (three)	PT Indosat Tbk (Matrix-Mentari-IM3)	PT Indosat Tbk (Matrix-Mentari-IM3)	EXCELCOM (XL)	EXCELCOM (XL)	TELKOMSEL (Halo-Simpati-AS)	EXCELCOM (XL)
DOWN LINK						
1945 - 1950 MHz	1950 - 1955 MHz	1955 - 1960 MHz	1960 - 1965 MHz	1965 - 1970 MHz	1970 - 1975 MHz	1975 - 1980 MHz
(Plan for second carrier)	(first carrier)	(second carrier)	(first carrier)	(second carrier)	(third carrier)	(third carrier)
UPLINK						

Floatway Learning Center

FLOATWAY SYSTEMS

3G Frequency Allocation in Indonesia

Frequency Spectrum Plan September 2013

PITA 3G				
Blok 1	Blok 2	Blok 3	Blok 4	Blok 5
2110 - 2115 Mhz	2115 - 2120 Mhz	2120 - 2125 MHz	2125 - 2130 MHz	2130 - 2135 MHz
PT Hutchison CP Telecommunications (three)	PT Hutchison CP Telecommunication (three)	TELKOMSEL (Halo-Simpati-AS)	TELKOMSEL (Halo-Simpati-AS)	TELKOMSEL (Halo-Simpati-AS)
DOWN LINK				
1920 - 1925 MHz	1925 - 1930 MHz	1930 - 1935 MHz	1935 - 1940 MHz	1940 - 1945 MHz
(first carrier)	(second carrier)	(first carrier)	(second carrier)	(third carrier)
UPLINK				

Spectrum Frequency

PITA 3G						
Blok 6	Blok 7	Blok 8	Blok 9	Blok 10	Blok 11	Blok 12
2135 - 2140 MHz	2140 - 2145 MHz	2145 - 2150 MHz	2150 - 2155 MHz	2155 - 2160 MHz	2160 - 2165 MHz	2165 - 2170 MHz
PT Indosat Tbk (Matrix-Mentari-IM3)	PT Indosat Tbk (Matrix-Mentari-IM3)	EXCELCOM (XL)	EXCELCOM (XL)	EXCELCOM (XL)	PT Natrindo Telepon Seluler (AXIS)	PT Natrindo Telepon Seluler (AXIS)
DOWN LINK						
1945 - 1950 MHz	1950 - 1955 MHz	1955 - 1960 MHz	1960 - 1965 MHz	1965 - 1970 MHz	1970 - 1975 MHz	1975 - 1980 MHz
(first carrier)	(second carrier)	(first carrier)	(second carrier)	(third carrier)	(first carrier)	(second carrier)
UPLINK						

Floatway Learning Center

FLOATWAY SYSTEMS

3G Frequency Allocation in Indonesia

Frequency Spectrum 2018

PITA 3G				
Blok 1	Blok 2	Blok 3	Blok 4	Blok 5
2110 - 2115 Mhz	2115 - 2120 Mhz	2120 - 2125 MHz	2125 - 2130 MHz	2130 - 2135 MHz
H3I (three)	H3I (three)	H3I (three)	TELKOMSEL (Halo-Simpati-AS)	TELKOMSEL (Halo-Simpati-AS)
DOWN LINK				
1920 - 1925 MHz	1925 - 1930 MHz	1930 - 1935 MHz	1935 - 1940 MHz	1940 - 1945 MHz
(first carrier)	(second carrier)	(third carrier)	(first carrier)	(second carrier)
UPLINK				

190 MHz Between Uplink-Downlink

PITA 3G						
Blok 6	Blok 7	Blok 8	Blok 9	Blok 10	Blok 11	Blok 12
2135 - 2140 MHz	2140 - 2145 MHz	2145 - 2150 MHz	2150 - 2155 MHz	2155 - 2160 MHz	2160 - 2165 MHz	2165 - 2170 MHz
TELKOMSEL (Halo-Simpati-AS)	XL	XL	XL	Indosat Ooredoo (Matrix-Mentari-IM3)	Indosat Ooredoo (Matrix-Mentari-IM3)	Indosat Ooredoo (Matrix-Mentari-IM3)
DOWN LINK						
1945 - 1950 MHz	1950 - 1955 MHz	1955 - 1960 MHz	1960 - 1965 MHz	1965 - 1970 MHz	1970 - 1975 MHz	1975 - 1980 MHz
(third carrier)	(first carrier)	(second carrier)	(third carrier)	(first carrier)	(second carrier)	(third carrier)
UPLINK						

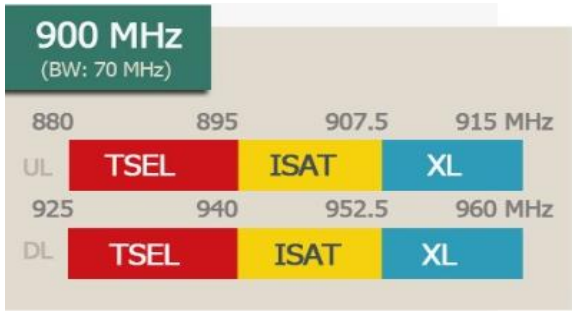
Floatway Learning Center

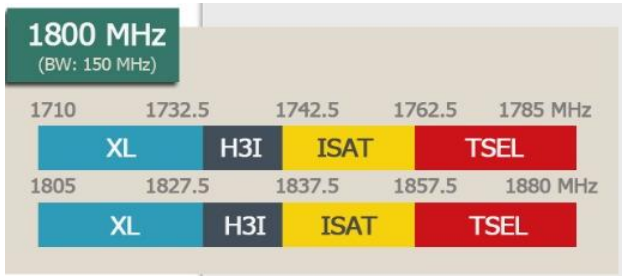
FLOATWAY SYSTEMS

Frekuensi Mobile Broadband in Indonesia













Floatway Learning Center

Floatway Learning Center

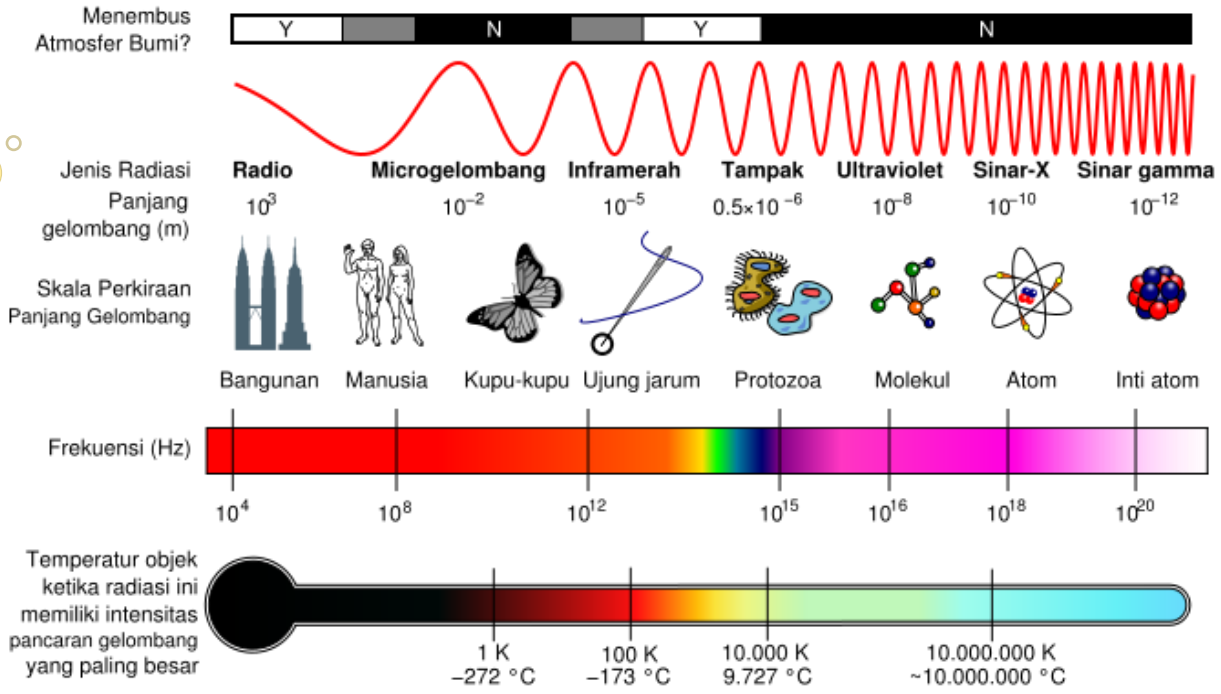
FDD IMT Frequency Bands

Band Number	Uplink (MHz)	Downlink (MHz)	Band Gap (MHz)	Duplex Separation (MHz)	UMTS Usage	LTE Usage
1	1920-1980	2110-2170	130	190	Y	Y
2	1850-1910	1930-1990	20	80	Y	Y
3	1710-1785	1805-1880	20	95	Y	Y
4	1710-1755	2110-2155	355	400	Y	Y
5	824-849	869-894	20	45	Y	Y
6	830-840	875-885	35	45	Y	Y
7	2500-2570	2620-2690	50	120	Y	Y
8	880-915	925-960	10	45	Y	Y
9	1749.9-1784.9	1844.9-1879.9	60	95	Y	Y
10	1710-1770	2110-2170	340	400	Y	Y
11	1427.9-1452.9	1475.9-1500.9	23	48	Y	Y
12	698-716	728-746	12	30	Y	Y
13	777-787	746-756	21	31	Y	Y
14	788-798	758-768	20	30	Y	Y
17	704-716	734-746	18	30	N	Y

TDD IMT Frequency Bands

Band	F_{low}-F_{high} (MHz)	UMTS	LTE
33	1900-1920	Y	Y
34	2010-2025	Y	Y
35	1850-1910	Y	Y
36	1930-1990	Y	Y
37	1910-1930	Y	Y
38	2570-2620	Y	Y
39	1880-1920	N	Y
40	2300-2400	Y	Y

Frequency & Wavelength



Panjang gelombang (λ) adalah jarak yang di ukur dari satu titik dari sebuah gelombang ke titik yang sama di gelombang selanjutnya.

λ= kecepatan cahaya / frekuensi (c = 3 x 10^{8} m/s)

Perilaku Gelombang Radio :

- Semakin panjang *panjang gelombang*, semakin jauh gelombang radio merambat.
- Semakin panjang *panjang gelombang*, semakin mudah gelombang radio melalui atau mengitari penghalang.
- Semakin pendek *panjang gelombang*, semakin banyak data yang dapat di kirim.

Floatway Learning Center

FLOATWAY SYSTEMS

Quiz 2 : 4G Frequency Allocation

Apabila operator-operator di bawah ini ingin migrasi ke jaringan LTE pada frequency berapa sajakah operator tersebut dapat mengimplementasikan jaringan LTE ?

Operator	Frequency LTE	Bandwidth	LTE TDD/FDD	Jaringan Sebelumnya
Bakrie Telecom				CDMA 800MHz
Hutchison				GSM1800, UMTS 2100
Indosat				GSM900, GSM1800, UMTS 2100
				CDMA 800MHz
Sampoerna Telekom				CDMA 450MHz
Smart Telecom				CDMA 1.900MHz
Telkom Flexi				CDMA 800MHz
Telkomsel				GSM900, GSM1800, UMTS 2100
XL Axiata				GSM900, GSM1800, UMTS 2100

Floatway Learning Center

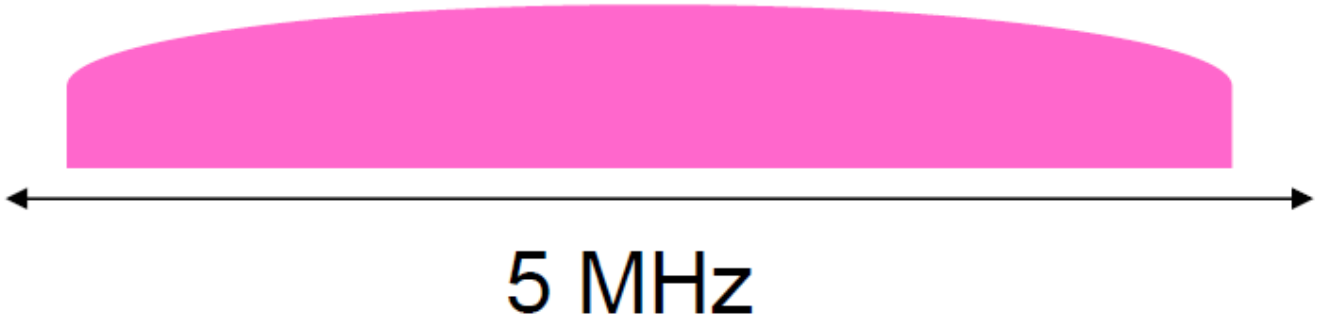
FLOATWAY SYSTEMS

Floatway Learning Center

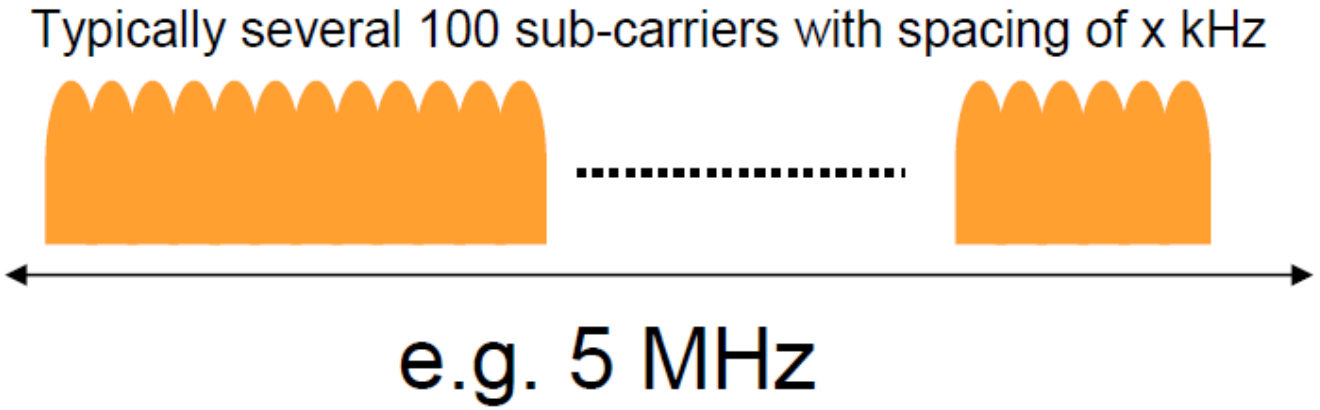
OFDMA

OFDM

- Single Carrier Transmission (e.g. WCDMA)



- Orthogonal Frequency Division Multiplexing



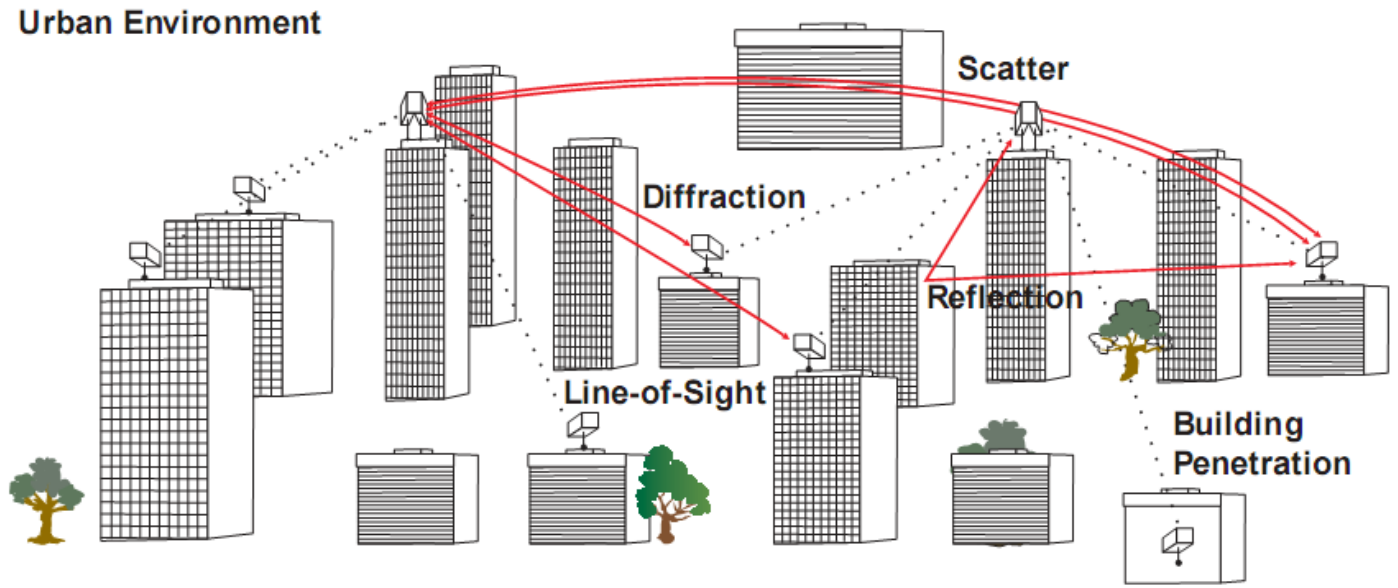
Floatway Learning Center

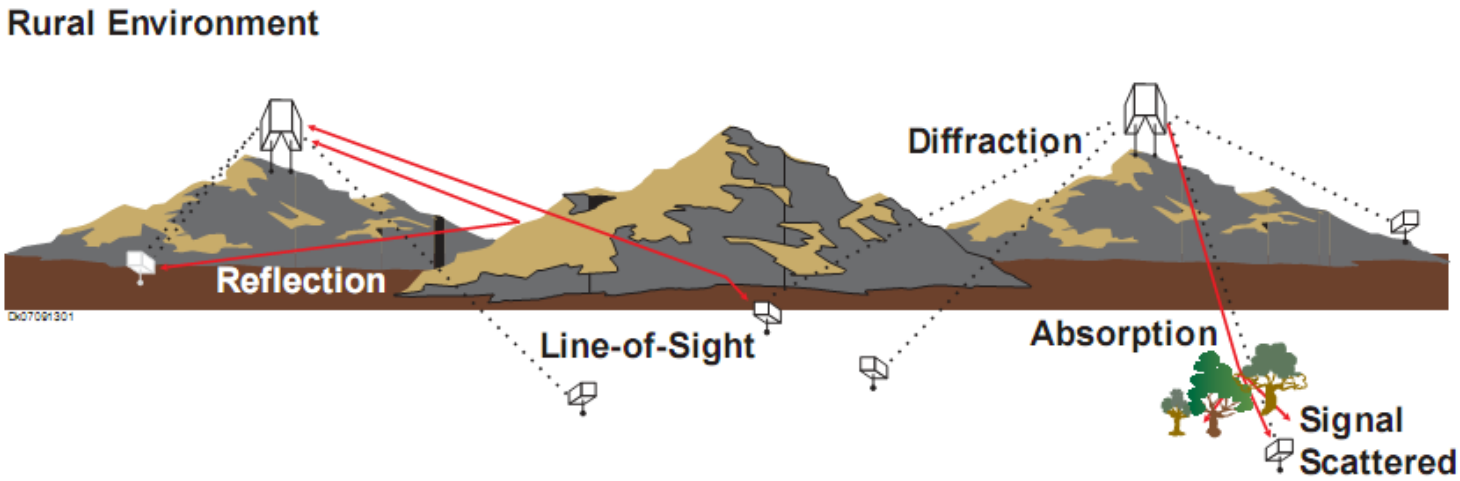
OFDM Concept: Mengapa OFDM

- Sinyal OFDM (Orthogonal Frequency Division Multiplexing) dapat mendukung kondisi NLOS (Non Line of Sight) dengan mempertahankan efisiensi spektral yang tinggi dan memaksimalkan spektrum yang tersedia.
- Mendukung lingkungan propagasi multi-path.
- Scalable bandwidth : menyediakan fleksibilitas dan potensial mengurangi CAPEX (capital expense).

Floatway Learning Center

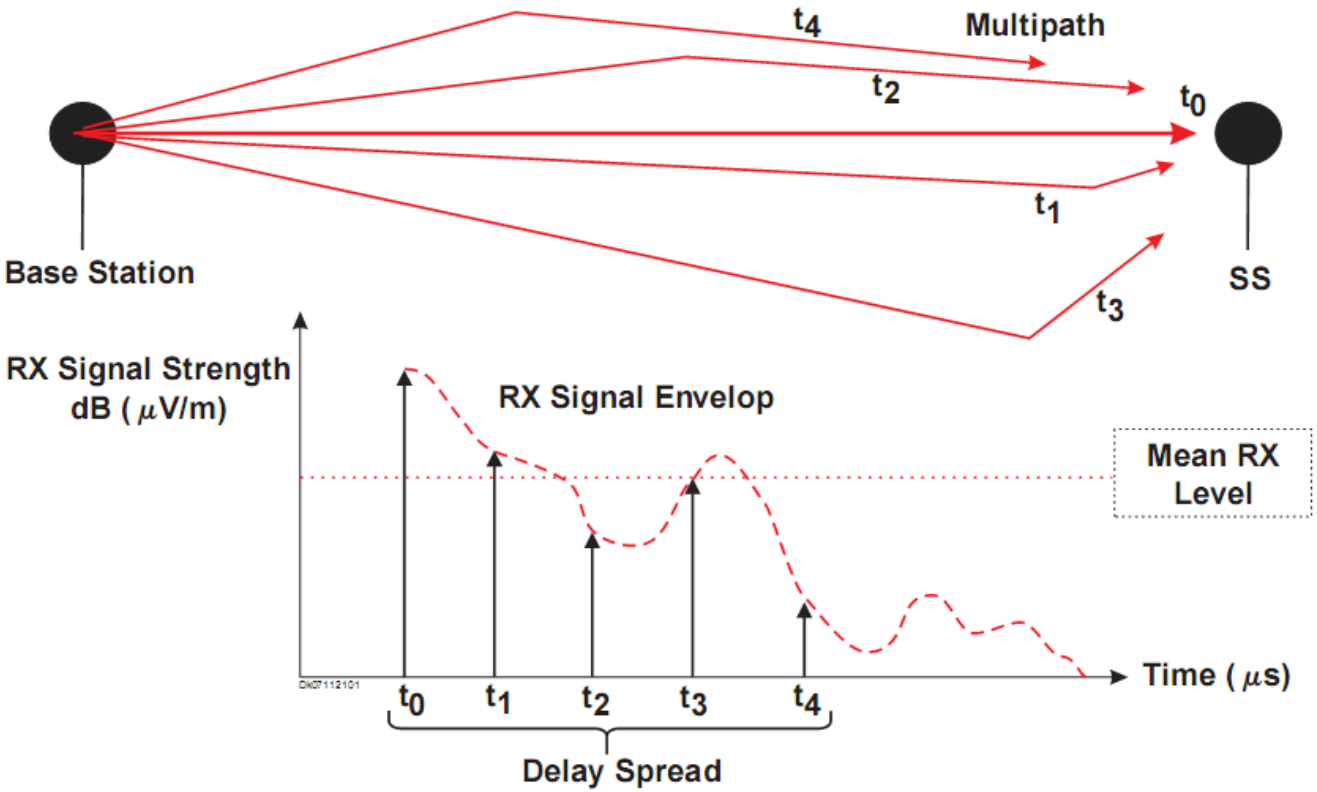
OFDM Concept: NLOS Performance





Floatway Learning Center

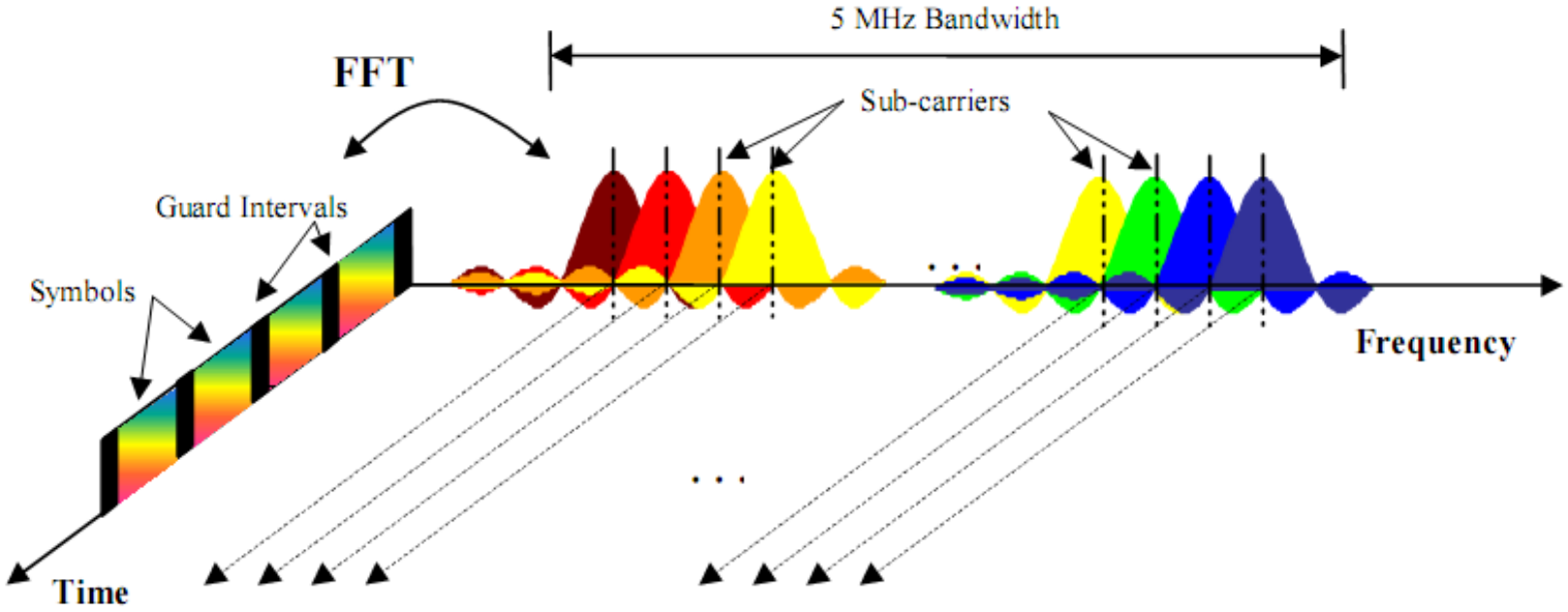
OFDM Concept : Mutipath Propagation



- Sinyal-sinyal multipath datang pada waktu yang berbeda dengan amplitudo dan pergeseran fasa yang berbeda, yang menyebabkan pelemahan dan penguatan daya sinyal yang diterima.
- Propagasi multipath berpengaruh terhadap performansi link dan coverage.
- Selubung (envelop) sinyal Rx berfluktuasi secara acak.

Floatway Learning Center

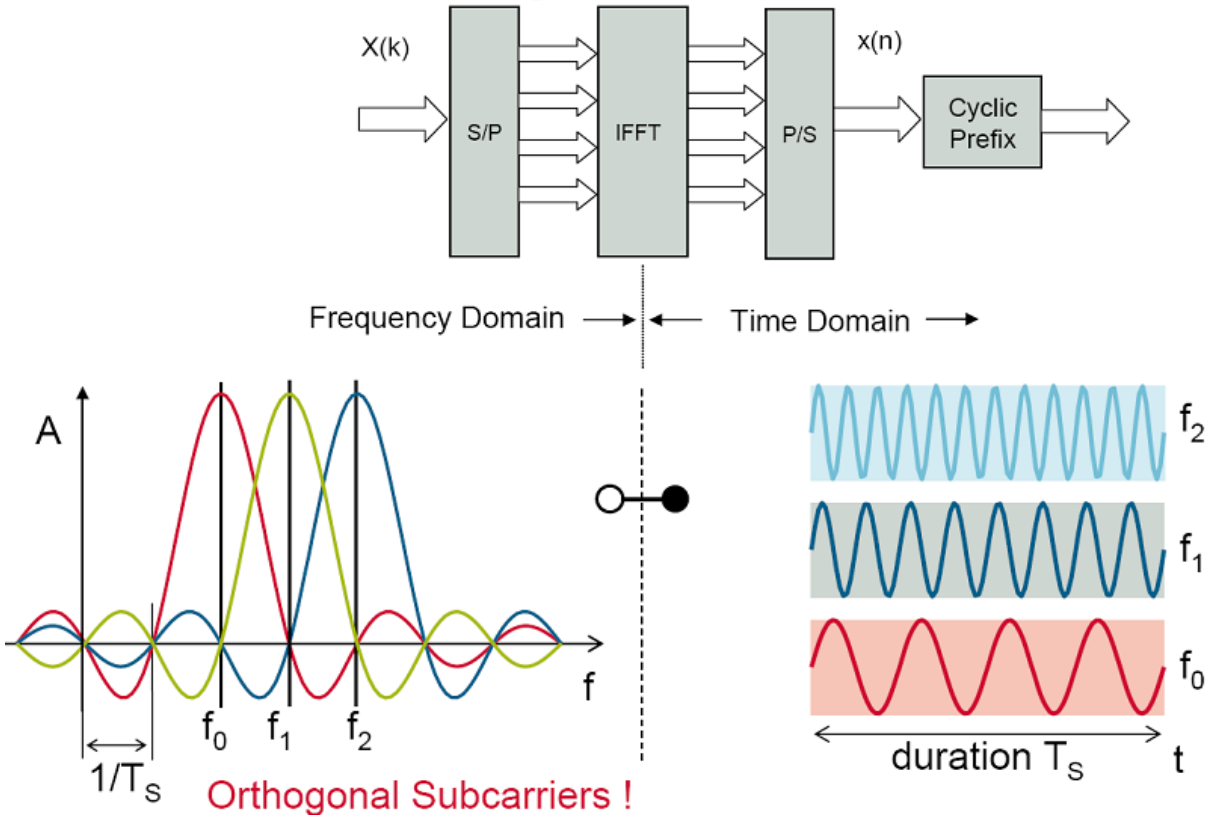
OFDM Concept: FFT



- Multi-carrier modulation/multiplexing technique
- Available bandwidth is divided into several subchannels
- Data is serial-to-parallel converted
- Symbols are transmitted on different subcarriers

Floatway Learning Center

OFDM Concept: IFFT

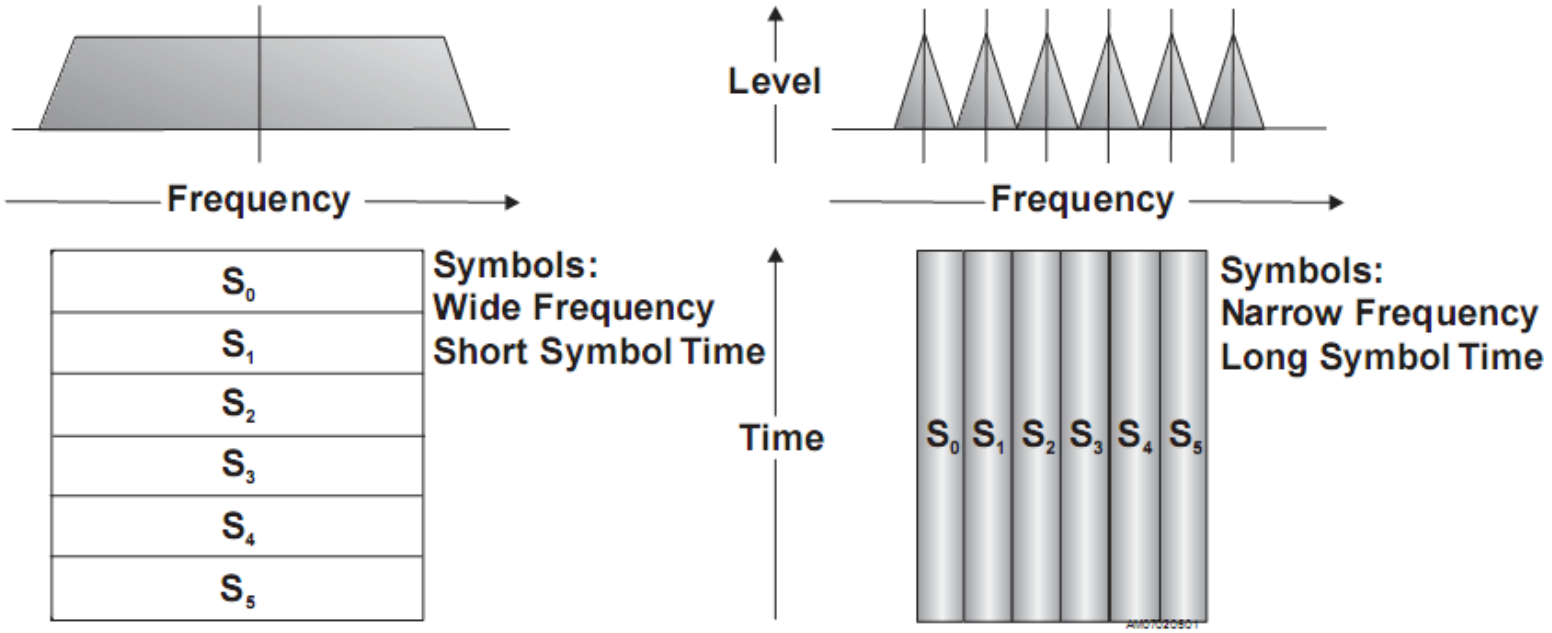


Basic ideas valid for various multicarrier techniques:

- OFDM: Orthogonal Frequency Division Multiplexing
- OFDMA: Orthogonal Frequency Division Multiple Access

Floatway Learning Center

OFDM Concept: Single-Carrier Vs. OFDM



Floatway Learning Center

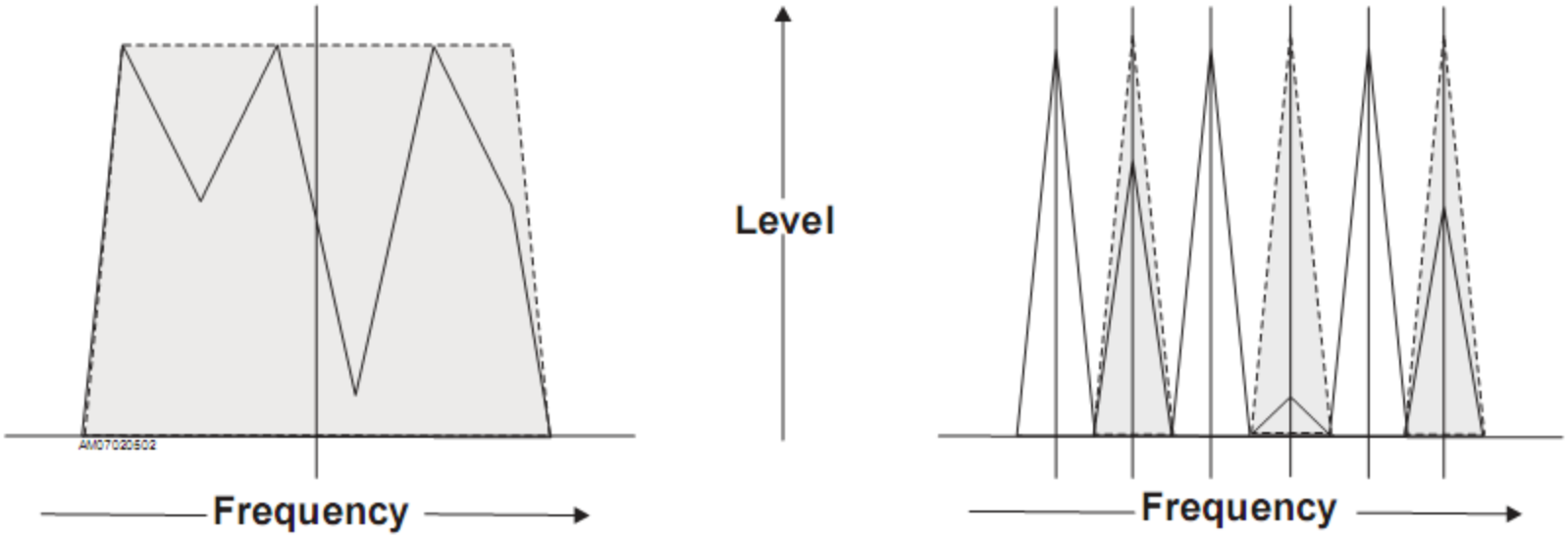
Single-Carrier Mode:

- Serial Symbol Stream Used to Modulate a Single Wideband Carrier
- Serial Datastream Converted to Symbols (Each Symbol Can Represented 1 or More Data Bits)

OFDM Mode:

- Each Symbol Used to Modulate a Separate Sub-Carrier

OFDM Concept: Single-Carrier Vs. OFDM



Single-Carrier Mode	OFDM Mode

- Dotted Area Represents Transmitted Spectrum
- Solid Area Represents Receiver Input

- OFDM mengatasi delay spread, multipath dan ISI (Inter Symbol Interference) secara efisien sehingga dapat meningkatkan throughput data rate yang lebih tinggi.
- Memudahkan ekualisasi kanal terhadap sub-carrier OFDM individual, dibandingkan terhadap sinyal single-carrier yang memerlukan teknik ekualisasi adaptif lebih kompleks.

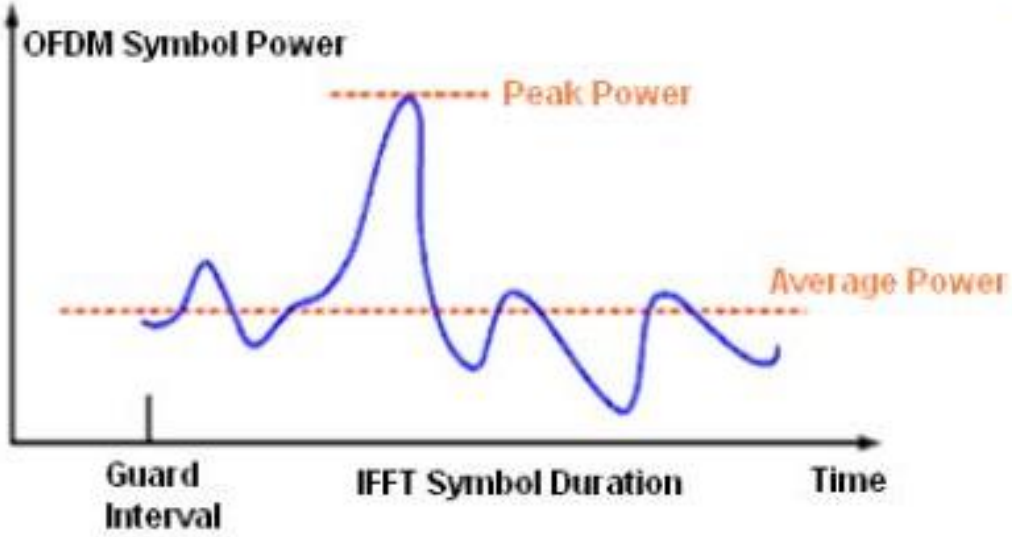
Floatway Learning Center

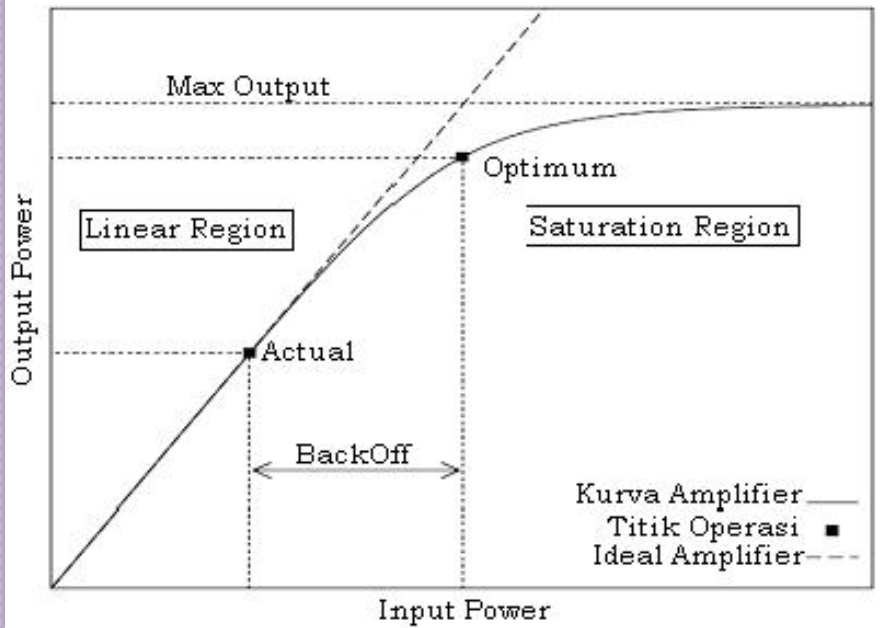
OFDM Concept: Motivation for Multi-carrier Approaches

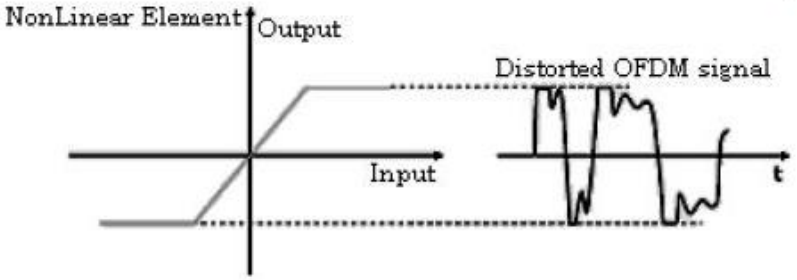
- Multi-carrier transmission offers various **advantages** over traditional single carrier approaches:
  - Highly scalable
  - Simplified equalizer design in the frequency domain, also in cases of large delay spread
  - High spectrum density
  - Simplified the usage of MIMO
- **Weakness** of multi-carrier systems:
  - Increased peak to average power ratio (PAPR)

Floatway Learning Center

OFDM Concept: Peak to Average Power Ratio (PAPR)

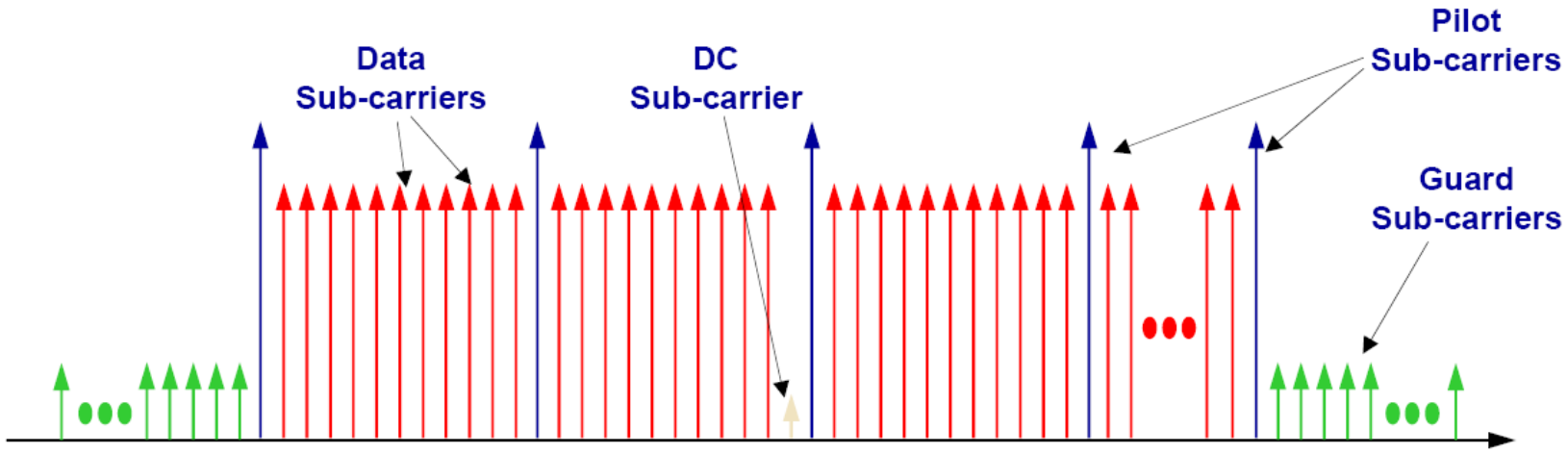






Floatway Learning Center

Tipe Sub-Carrier OFDM



Data Sub-carriers

- Membawa simbol BPSK, QPSK, 16QAM, 64QAM

Pilot Sub-carriers

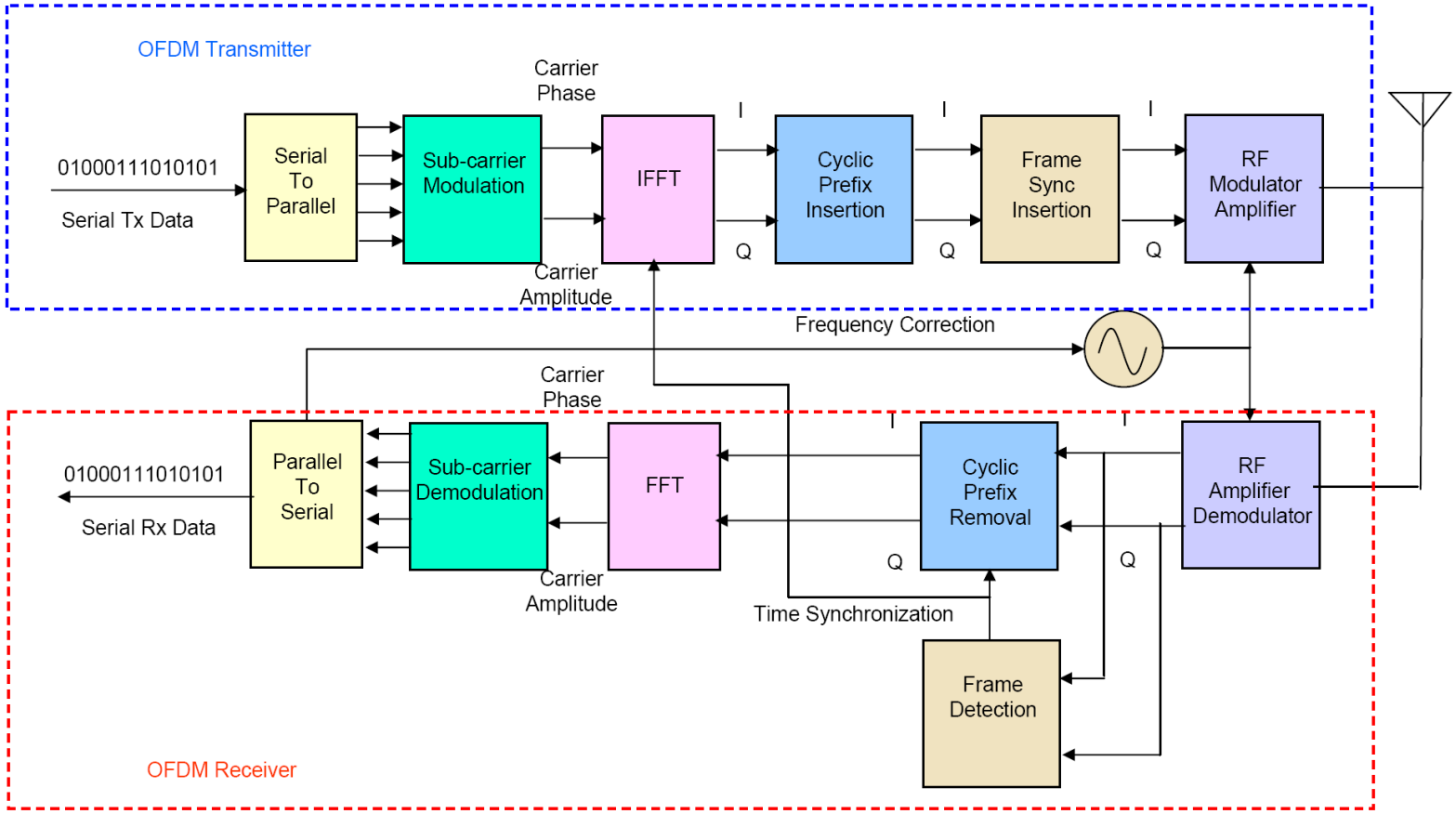
- Untuk memudahkan estimasi kanal dan demodulasi koheren pada receiver.

Null Subcarrier

- Guard Sub-carriers
- DC Sub-carrier

Floatway Learning Center

OFDM Transceiver



Floatway Learning Center

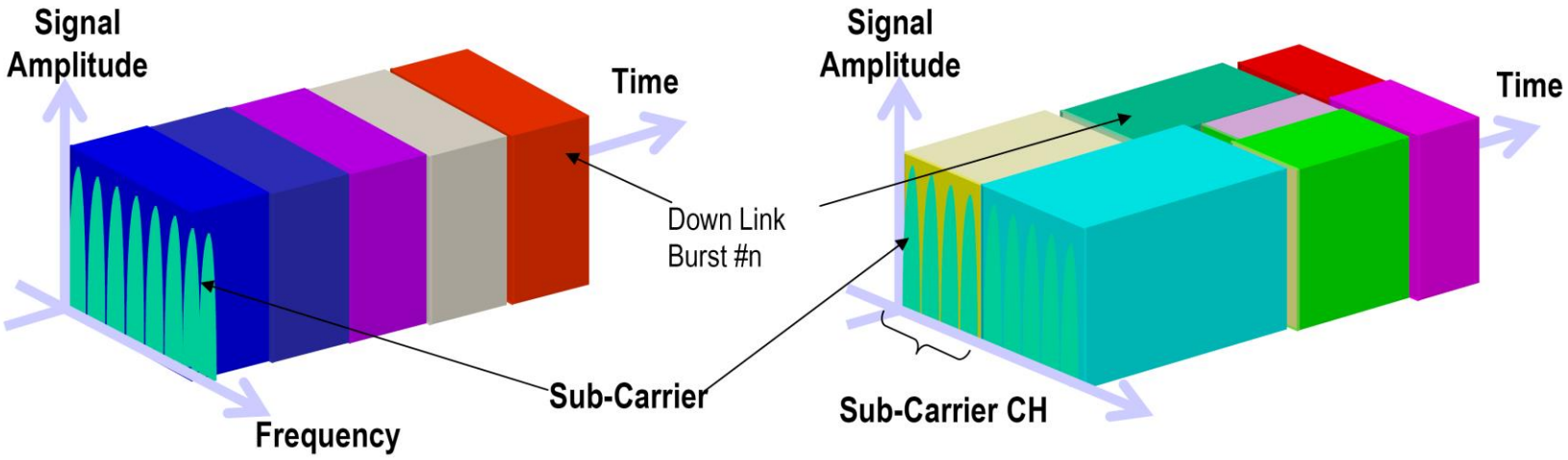
OFDM & OFDMA

OFDM

- Semua subcarrier dialokasikan untuk satu user
- Misal : 802.16-2004

OFDMA

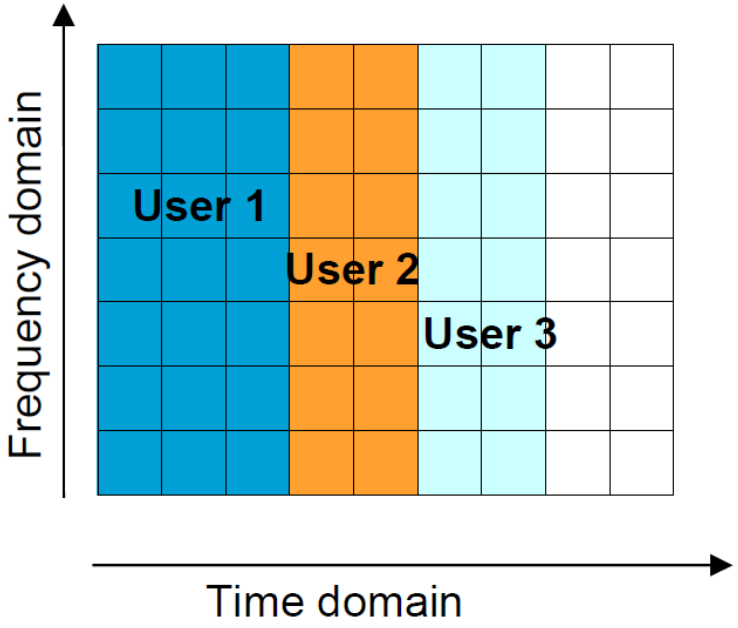
- Subcarrier dialokasikan secara fleksibel untuk banyak user tergantung pada kondisi radio.
- Misal : 802.16e-2005 dan 802.16m

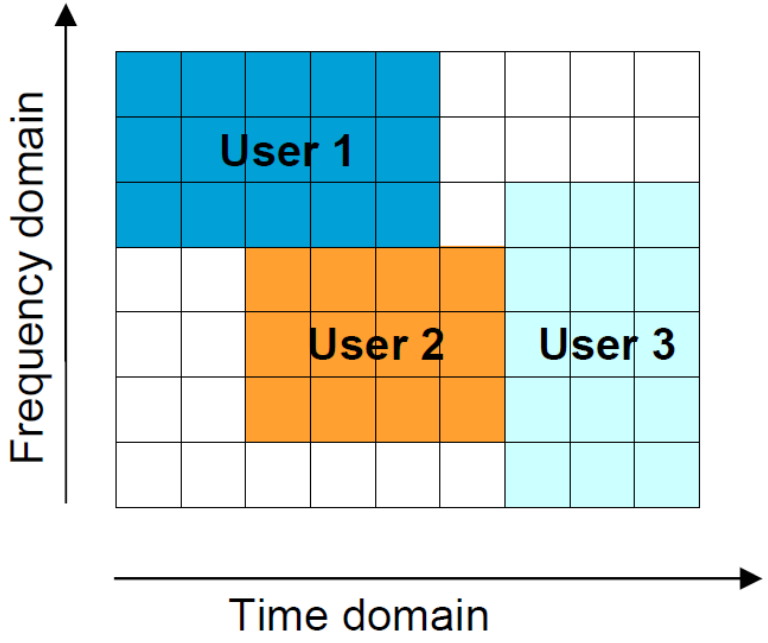


Floatway Learning Center

Difference between OFDM and OFDMA

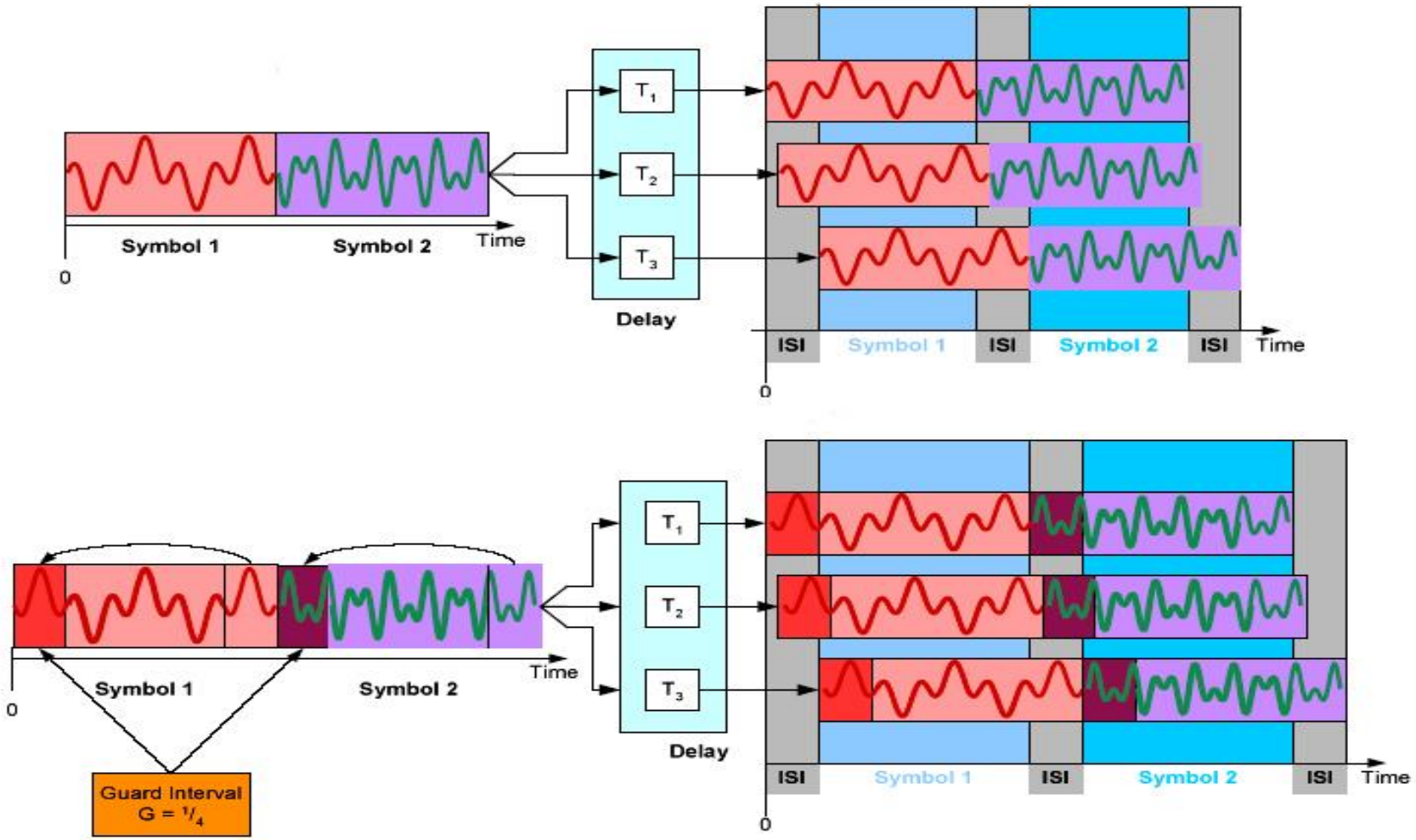
- OFDM allocates users in time domain only
- OFDMA allocates users in time and frequency domain





Floatway Learning Center

Guard Interval (Cyclic Prefix)



- Untuk mengatasi multipath delay spread
- Contoh pada WiMAX Guard Interval (cyclic prefix) : 1/4, 1/8, 1/16 or 1/32

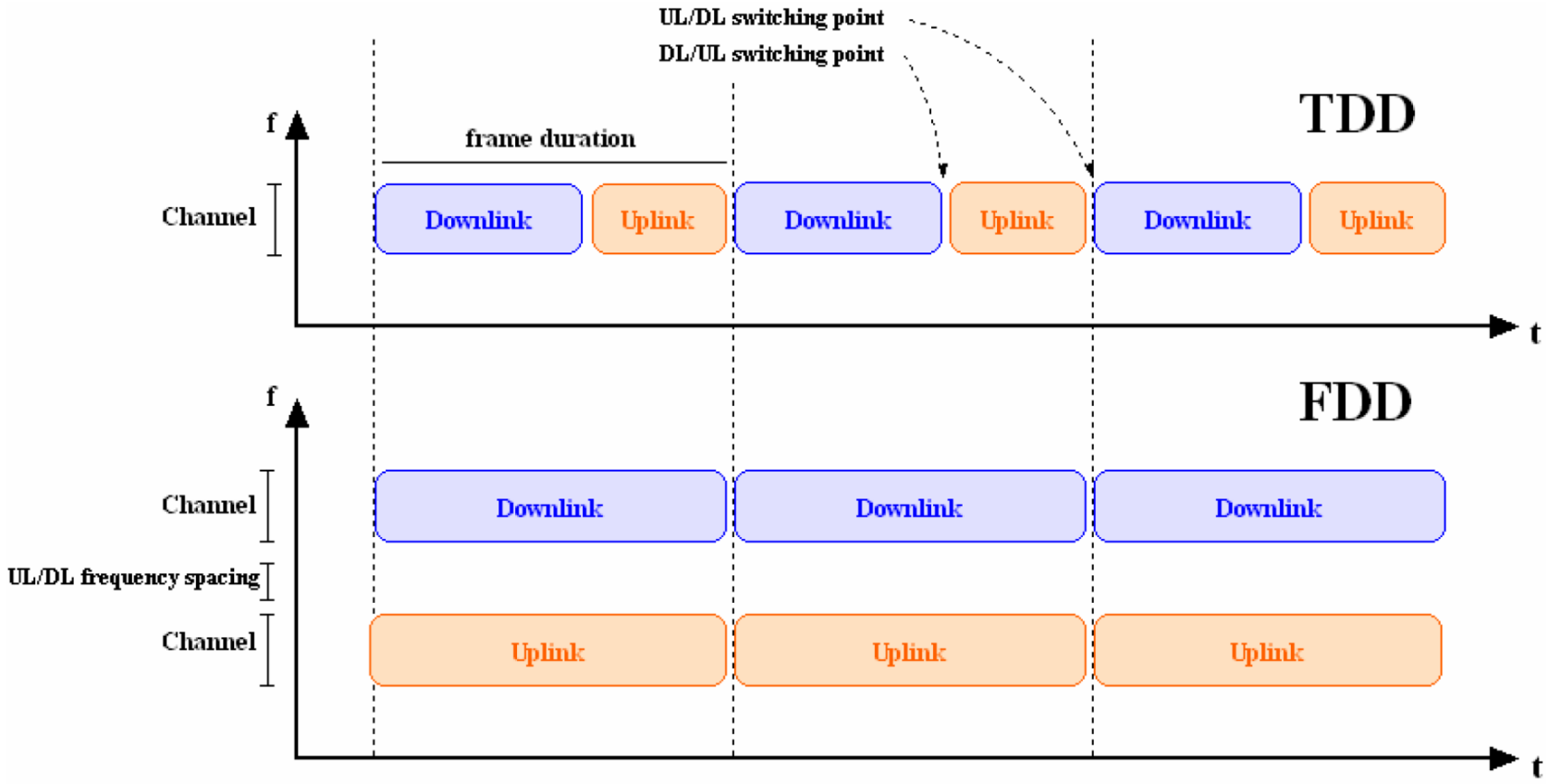
Floatway Learning Center

Cyclic Prefix Configuration

Configuration	Number of symbols N_{symb}^{DL}	Cyclic Prefix length in samples	Cyclic Prefix length in μs
Normal cyclic prefix Δf=15 kHz	7	160 for first symbol 144 for other symbols	5.2 μs for first symbol 4.7 μs for other symbols
Extended cyclic prefix Δf=15 kHz	6	512	16.7 μs
Extended cyclic prefix Δf=7.5 kHz	3	1024	33.3 μs

Floatway Learning Center

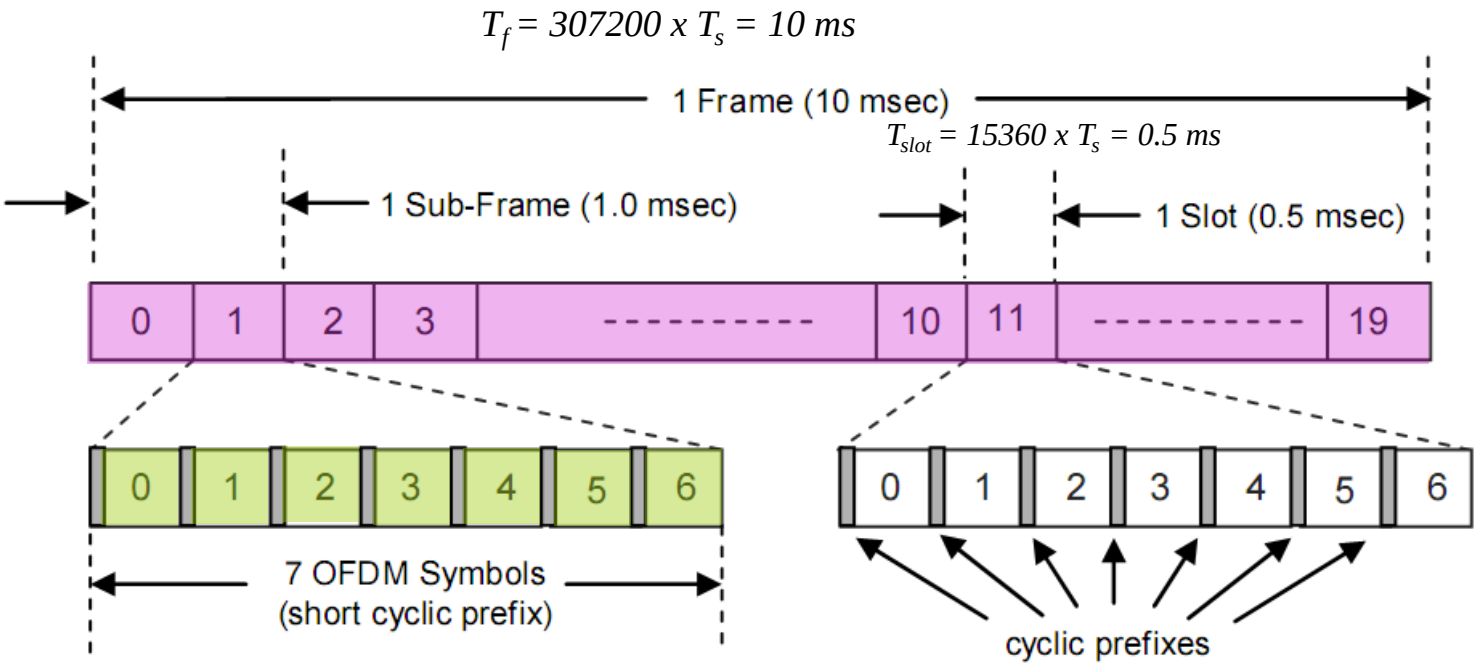
TDD & FDD



- Time Division Duplex (TDD)
- Frequency Division Duplex (FDD)
- Durasi Frame : 2.5 - 20ms

Floatway Learning Center

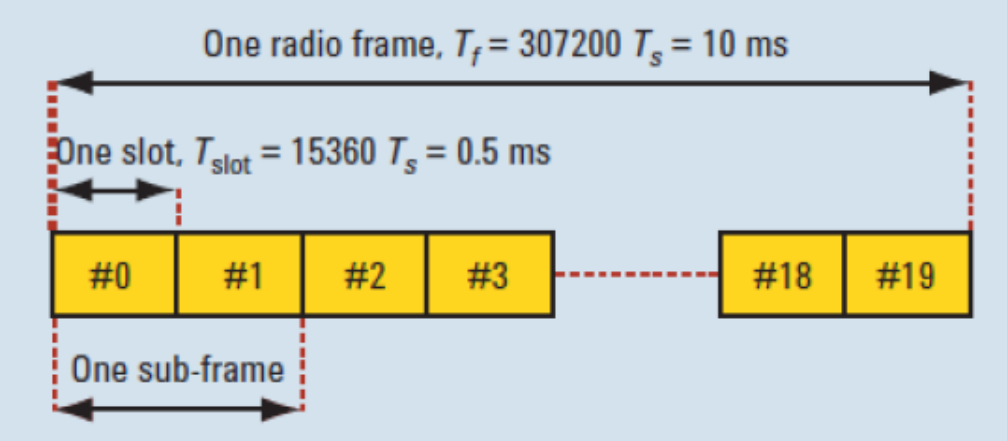
Generic LTE Frame Structure type 1 (FDD)



- Untuk struktur generik, frame radio 10 ms dibagi dalam 20 slot yang sama berukuran 0.5 ms.
- Suatu sub-frame terdiri dari 2 slot berturut-turut, sehingga satu frame radio berisi 10 sub-frame.
- Ts menunjukkan unit waktu dasar yang sesuai dengan 30.72 MHz.
- Struktur frame tipe-1 dapat digunakan untuk transmisi FDD dan TDD.

Floatway Learning Center

LTE Frame Structure type 1 (FDD)

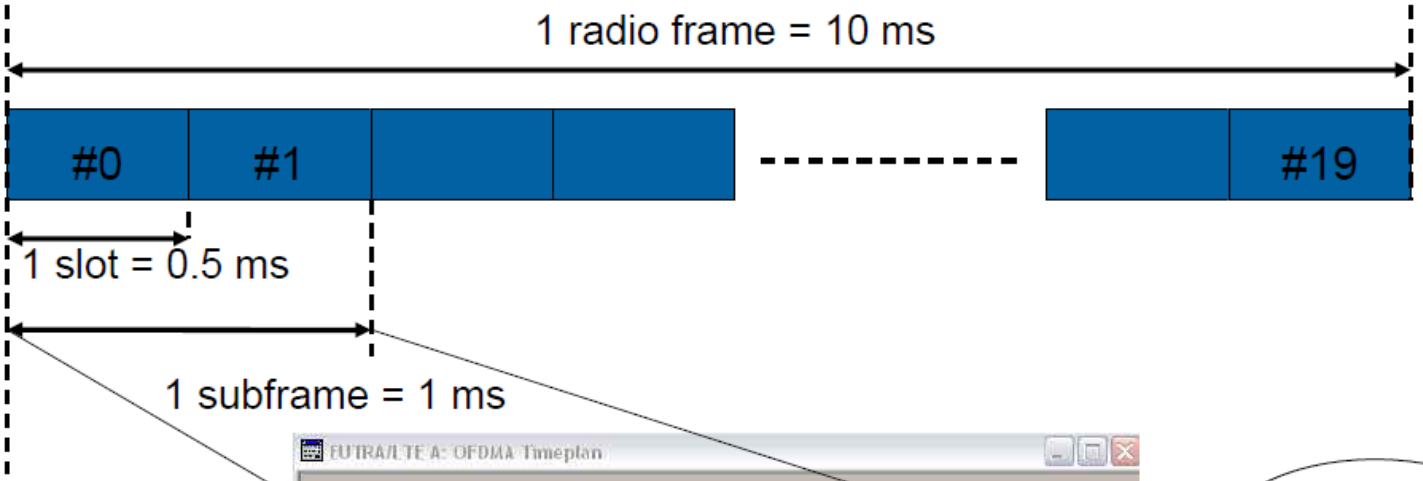


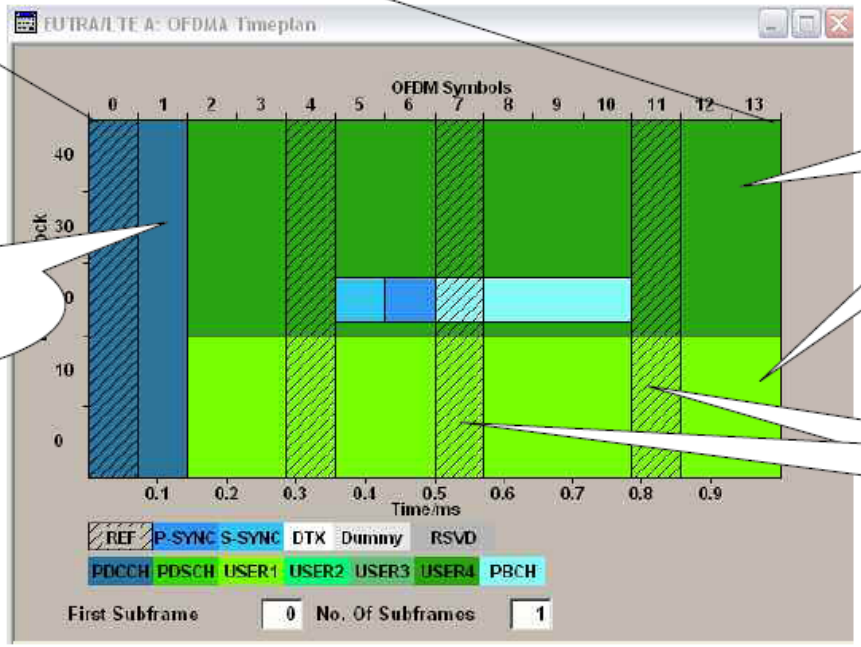
Description	Name	Value
Radio Frame length	T_f	10 ms
Subframe length	T_{sf}	1 ms
Slot length	T_s	0.5 ms
Subcarrier spacing	Δf	15 kHz (constant)
Sample frequency (time unit)	T_s	1/(15k*2048) = 32.6 ns
OFDMA Symbol length		~ 70 us
Resource block		12 subcarriers

- 2 slots form one subframe = 1 ms
- For FDD, in each 10 ms interval, all 10 subframes are available for downlink transmission and uplink transmissions.
- For TDD, a subframe is either located to downlink or uplink transmission. The 0th and 5th subframe in a radio frame is always allocated for downlink transmission.

Floatway Learning Center

Downlink LTE Frame Structure type 1 (FDD)





L1/2 downlink control channels

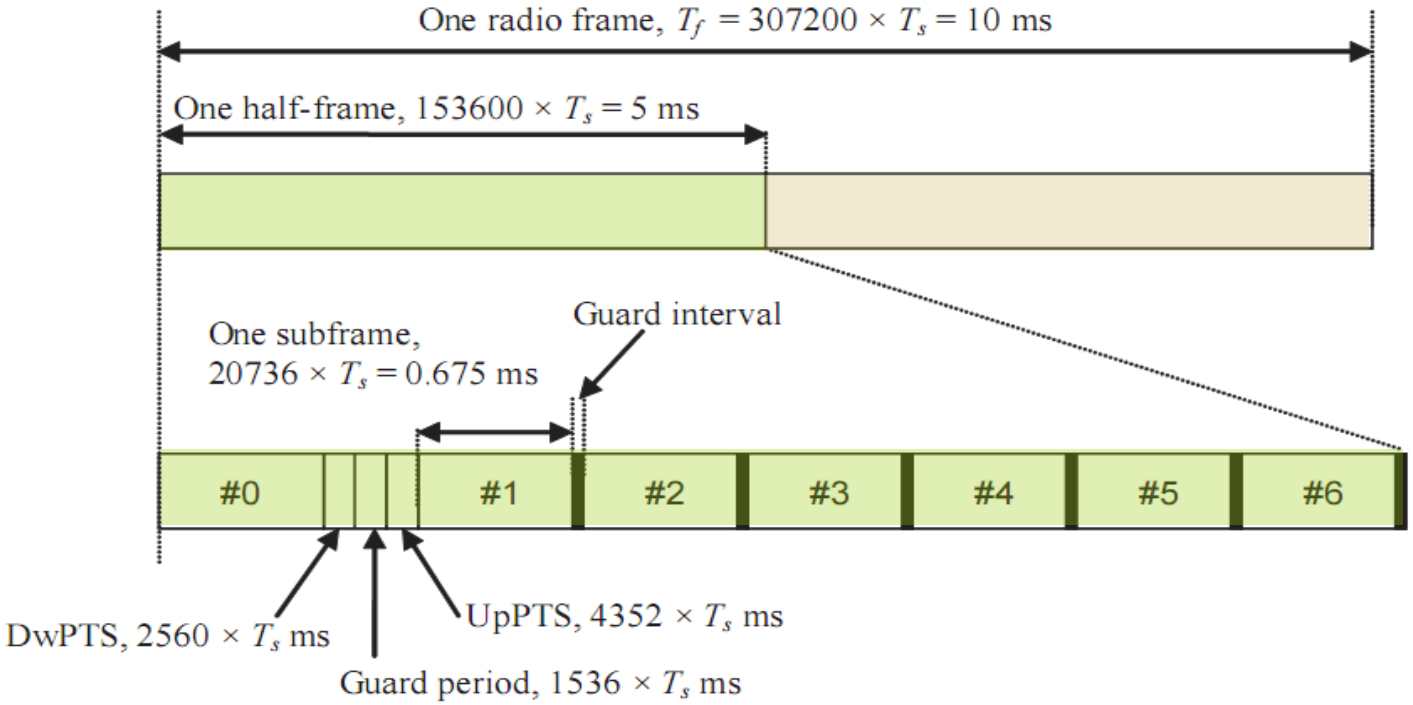
User data allocations

Downlink reference signal

Screenshot of R&S SMU200A signal generator

Floatway Learning Center

Generic LTE Frame Structure type 2 (TDD)

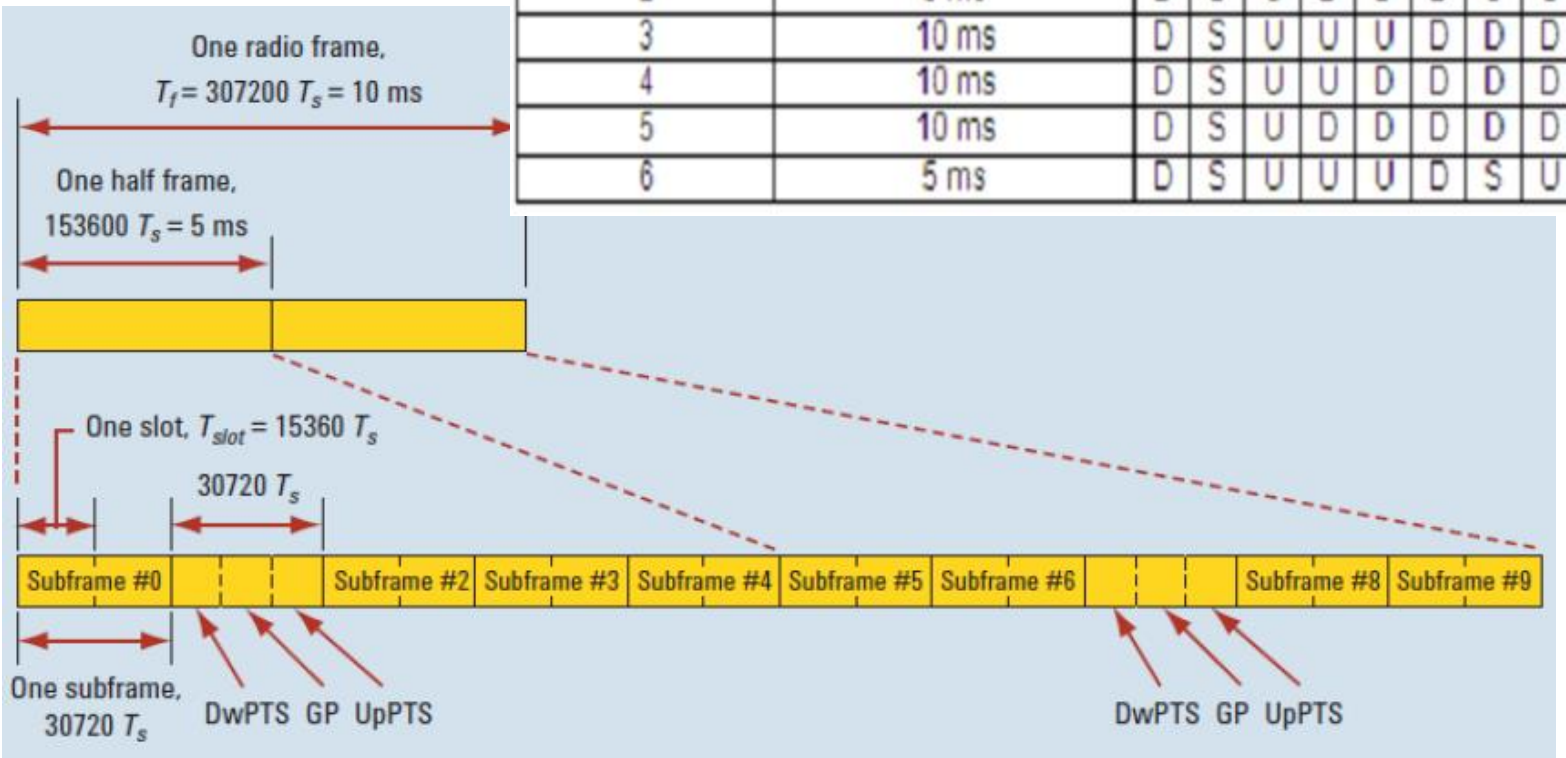


- Struktur frame tipe-2 hanya digunakan untuk transmisi TDD.
- Slot 0 dan DwPTSdisediakan untuk transmisi DL, sedangkan slot 1 dan UpPTS disediakan untuk transmisi UL.

Floatway Learning Center

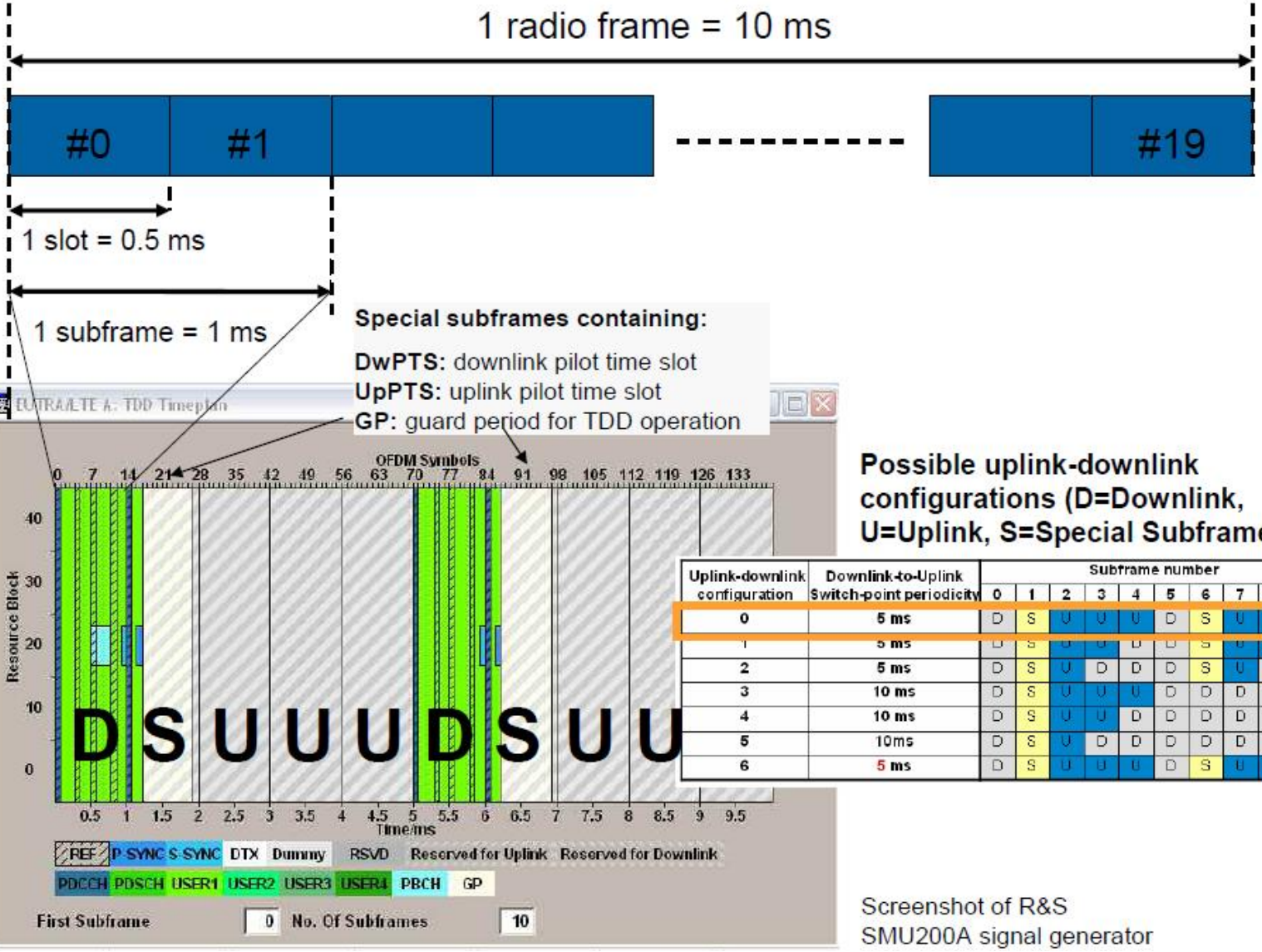
LTE Frame Structure type 2 (TDD)

Uplink-downlink configuration	Downlink-to-Uplink Switch-point periodicity	Subframe number									
		0	1	2	3	4	5	6	7	8	9
0	5 ms	D	S	U	U	U	D	S	U	U	U
1	5 ms	D	S	U	U	D	D	S	U	U	D
2	5 ms	D	S	U	D	D	D	S	U	D	D
3	10 ms	D	S	U	U	U	D	D	D	D	D
4	10 ms	D	S	U	U	D	D	D	D	D	D
5	10 ms	D	S	U	D	D	D	D	D	D	D
6	5 ms	D	S	U	U	U	D	S	U	U	D



Floatway Learning Center

LTE Frame Structure type 2 (TDD)



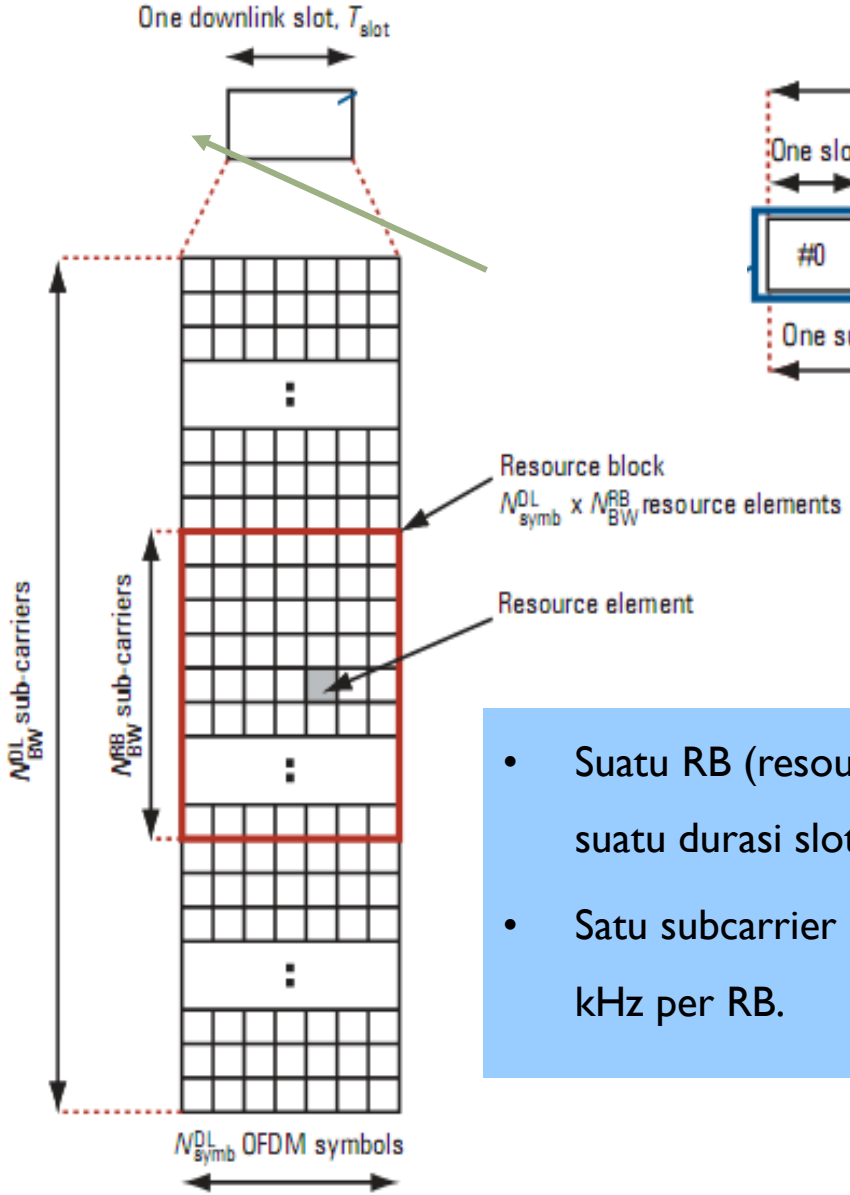
Possible uplink-downlink configurations (D=Downlink, U=Uplink, S=Special Subframe):

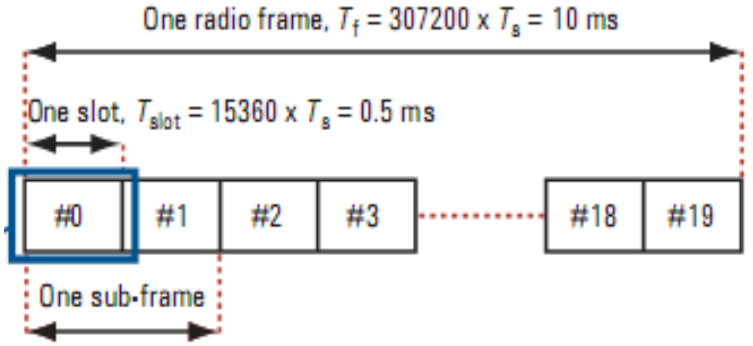
Uplink-downlink configuration	Downlink-to-Uplink Switch-point periodicity	Subframe number									
		0	1	2	3	4	5	6	7	8	9
0	5 ms	D	S	U	U	U	D	S	U	U	U
1	5 ms	D	S	U	U	D	D	S	U	U	D
2	5 ms	D	S	U	D	D	D	S	U	D	D
3	10 ms	D	S	U	U	U	D	D	D	D	D
4	10 ms	D	S	U	U	D	D	D	D	D	D
5	10ms	D	S	U	D	D	D	D	D	D	D
6	5 ms	D	S	U	U	U	D	S	U	U	D

Screenshot of R&S SMU200A signal generator

Floatway Learning Center

LTE Radio Frame - Sub Frame - Slot





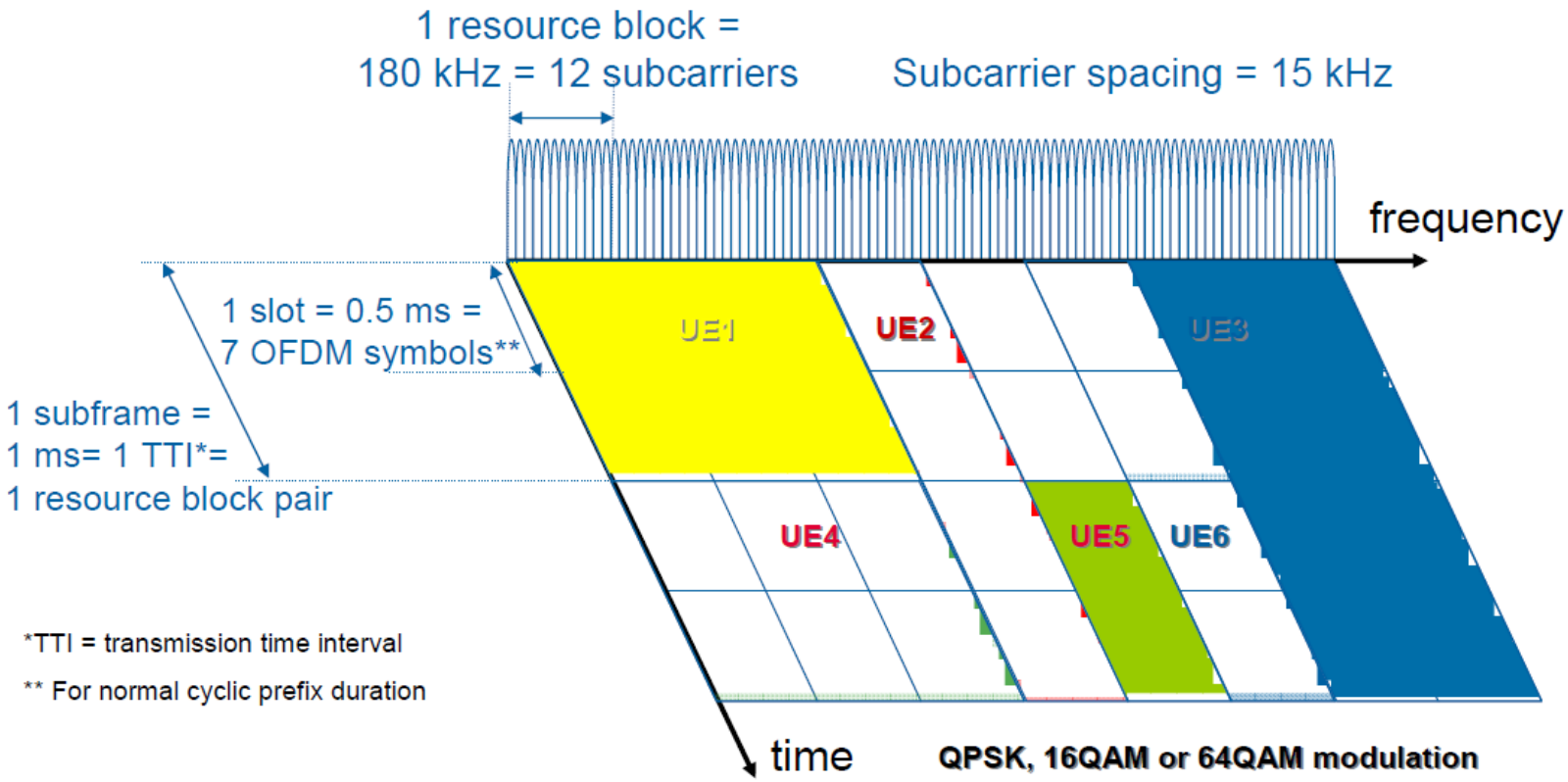
7 symbols X 12 subcarriers (short CP), or;

6 symbols X 12 subcarriers (long CP)

- Suatu RB (resource block) terdiri dari 12 subcarrier pada suatu durasi slot 0.5 ms.
- Satu subcarrier mempunyai BW 15 kHz, sehingga menjadi 180 kHz per RB.

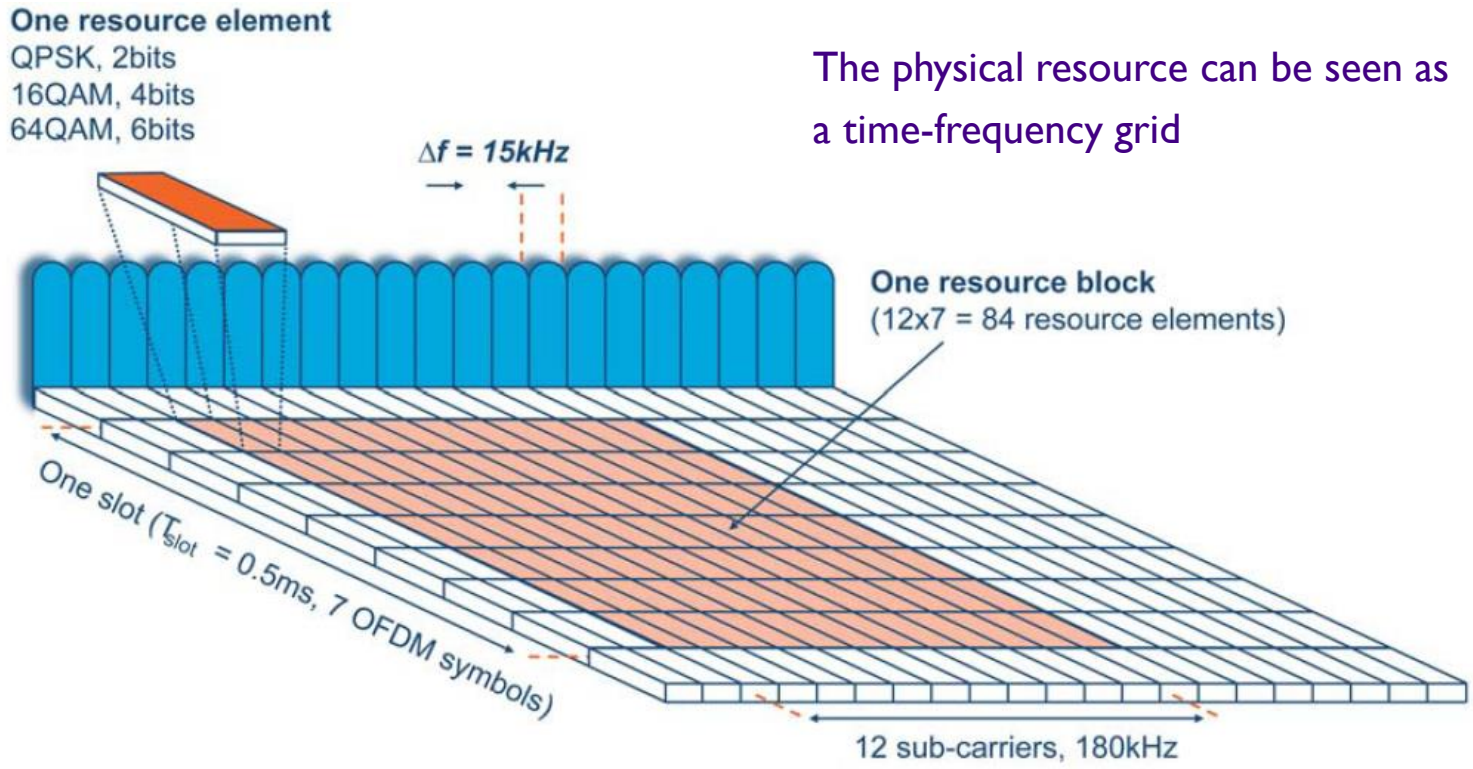
Floatway Learning Center

OFDMA time-frequency multiplexing



Floatway Learning Center

LTE Downlink Physical Layer Design: Physical Resource



The physical resource can be seen as a time-frequency grid

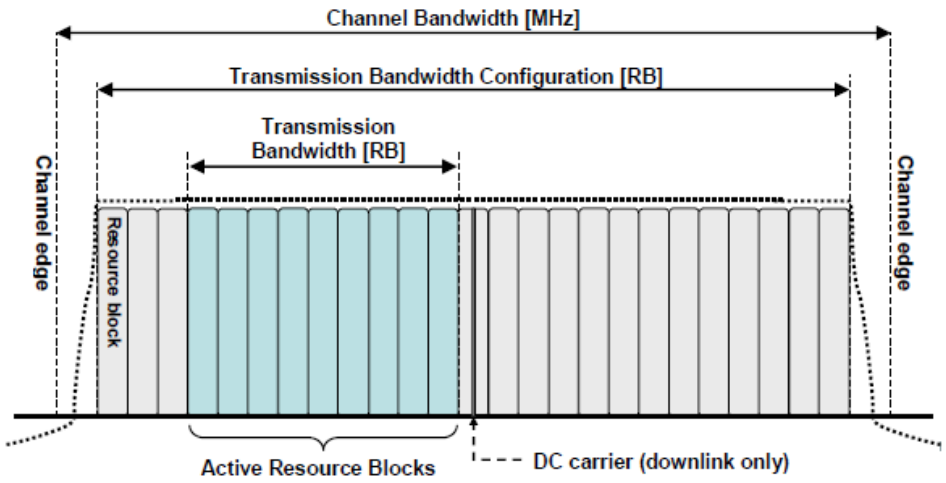
- LTE uses OFDM (Orthogonal Frequency Division Multiplexing) as its radio technology in downlink
- In the uplink LTE uses a pre=coded version of OFDM, SC-FDMA (Single Carrier Frequency Division Multiple Access) to reduced power consumption

Floatway Learning Center

LTE – Spectrum Flexibility

- LTE physical layer supports any bandwidth from 1.4 MHz to 20 MHz in steps of 180 kHz (resource block).
- Current LTE specification supports a subset of 6 different system bandwidths.
- All UEs must support the maximum bandwidth of 20 MHz.

Channel bandwidth $BW_{Channel}$ [MHz]	1.4	3	5	10	15	20
Number of resource blocks	6	15	25	50	75	100



Floatway Learning Center

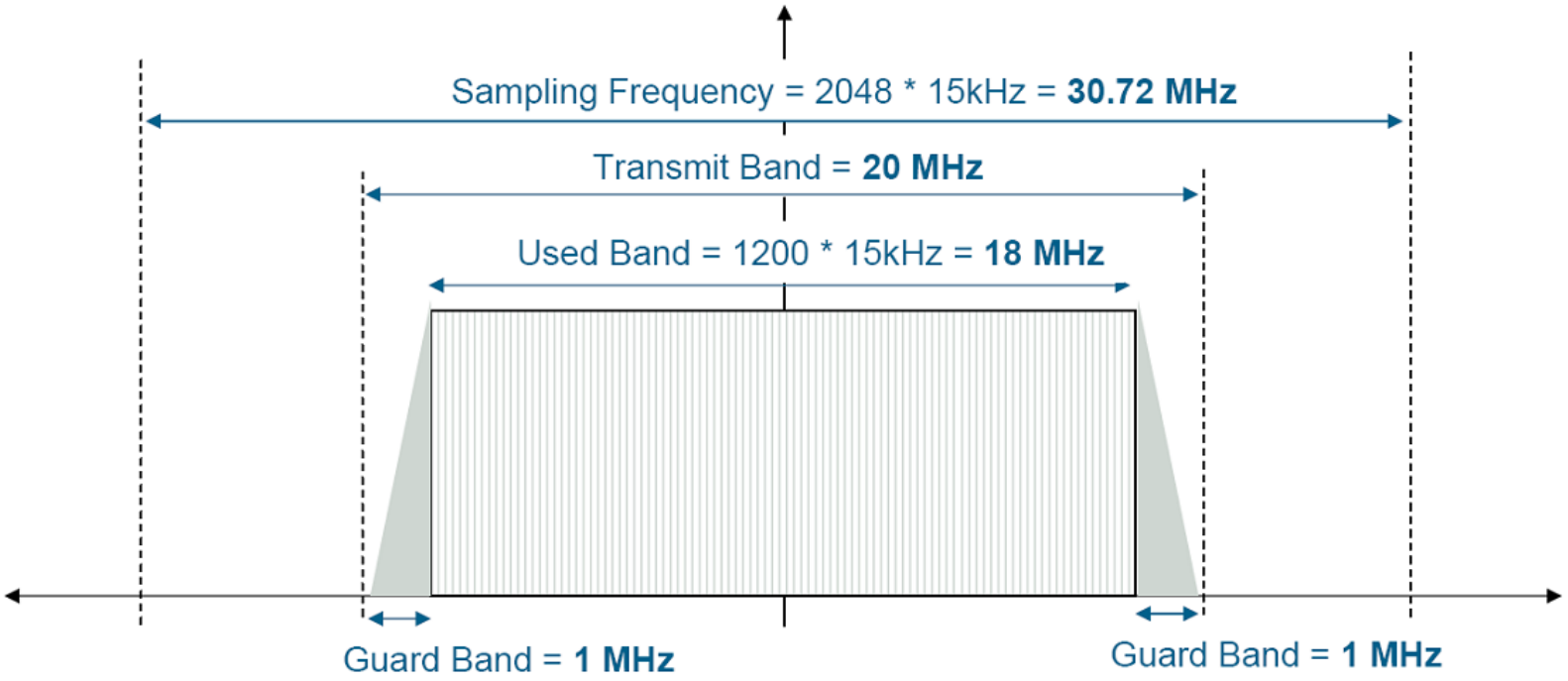
LTE – Spectrum Flexibility

Transmission BW		1.25 MHz	2.5 MHz	5 MHz	10 MHz	15 MHz	20 MHz
Sub-frame duration		0.5 ms					
Sub-carrier spacing		15 kHz					
Sampling frequency		1.92 MHz (1/2x3.84 MHz)	3.84 MHz	7.68 MHz (2x3.84 MHz)	15.36 MHz (4x3.84 MHz)	23.04 MHz (6x3.84 MHz)	30.72 MHz (8x3.84 MHz)
FFT size		128	256	512	1024	1536	2048
OFDM sym per slot (short/long CP)		7/6					
CP length (usec/ samples)	Short	(4.69/9) x 6, (5.21/10) x 1	(4.69/18) x 6, (5.21/20) x 1	(4.69/36) x 6, (5.21/40) x 1	(4.69/72) x 6, (5.21/80) x 1	(4.69/108) x 6, (5.21/120) x 1	(4.69/144) x 6, (5.21/160) x 1
	Long	(16.67/32)	(16.67/64)	(16.67/128)	(16.67/256)	(16.67/384)	(16.67/512)

Floatway Learning Center

Case Study

LTE Signal Spectrum (20 MHz case)



- The LTE standard uses an over-sized LTE. The actual used bandwidth is controlled by the number of used subcarriers. 15 kHz subcarrier spacing is the constant factor!
- 18 MHz out of 20 MHz is used for data, 1 MHz on each side is used as guard band.
- LTE used spectrum radio = 90%
- WiMAX used spectrum radio = 82%

Floatway Learning Center

DL Peak rates for E-UTRA FDD/TDD frame structure type 1

	Downlink	
Assumptions	64 QAM Signal overhead for reference signals and control channel occupying one OFDM symbol	
Unit	Mbps in 20 MHz	b/s/Hz
Requirement	100	5.0
2x2 MIMO	172.8	8.6
4x4 MIMO	326.4	16.3

3GPP TR 25.912 V11.0.0 (2012-09)

Technical Report

3rd Generation Partnership Project;
Technical Specification Group Radio Access Network;
Feasibility study for
evolved Universal Terrestrial Radio Access (UTRA)
and Universal Terrestrial Radio Access Network (UTRAN)
(Release 11)

Floatway Learning Center

UL Peak rates for E-UTRA FDD/TDD frame structure type 1

	Uplink	
Assumptions	Single TX UE Signal overhead for reference signals and control channel occupying 2RB	
Unit	Mbps in 20 MHz	b/s/Hz
Requirement	50	2.5
16QAM	57.6	2.9
64QAM	86.4	4.3

3GPP TR 25.912 V11.0.0 (2012-09)

Technical Report

3rd Generation Partnership Project;
Technical Specification Group Radio Access Network;
Feasibility study for
evolved Universal Terrestrial Radio Access (UTRA)
and Universal Terrestrial Radio Access Network (UTRAN)
(Release 11)

Floatway Learning Center

Peak rates for E-UTRA TDD
frame structure type 2

	Downlink		Uplink	
Assumptions	64 QAM, R=1		Single TX UE, 64 QAM, R=1	
Unit	Mbps in 20 MHz	b/s/Hz	Mbps in 20 MHz	b/s/Hz
Requirement	100	5.0	50	2.5
2x2 MIMO in DL	142	7.1	62.7	3.1
4x4 MIMO in DL	270	13.5		

3GPP TR 25.912 V11.0.0 (2012-09)

Technical Report

3rd Generation Partnership Project;
Technical Specification Group Radio Access Network;
Feasibility study for
evolved Universal Terrestrial Radio Access (UTRA)
and Universal Terrestrial Radio Access Network (UTRAN)
(Release 11)

Floatway Learning Center

Quiz 3 : LTE DL Peak Rate

Hitunglah maksimum Peak untuk cell eNodeB dengan Bandwith 1.4 MHz dengan modulasi QPSK, normal/short CP

14 OFDM symbols per 1 ms subframe
QPSK = ... bits per symbol
..... X ..... = ... bits per 1 ms subframe

..... bits / ..... ms = ... Kbps per subcarrier
..... X ..... kbps = ... Mbps per Resource Block
..... Resource Block in ..... Mhz
..... X ..... Mbps = ... Mbps per antenna

Floatway Learning Center

Quiz 4 : LTE DL Peak Rate

Hitunglah maksimum Peak Rate dari masing-masing Konfigurasi Lte dibawah ini dengan menggunakan normal/short CP.

$$PeakBitRate[Mbps] = \frac{bit}{Hz} xN_{subcarriers} x \frac{N_{symbol-per-subframe}}{1ms}$$

Bandwidth (MHz)	Modulation		
	QPSK	16 QAM	64 QAM
1.4			
3			
5			
10			
15			
20			

Floatway Learning Center

Floatway Learning Center

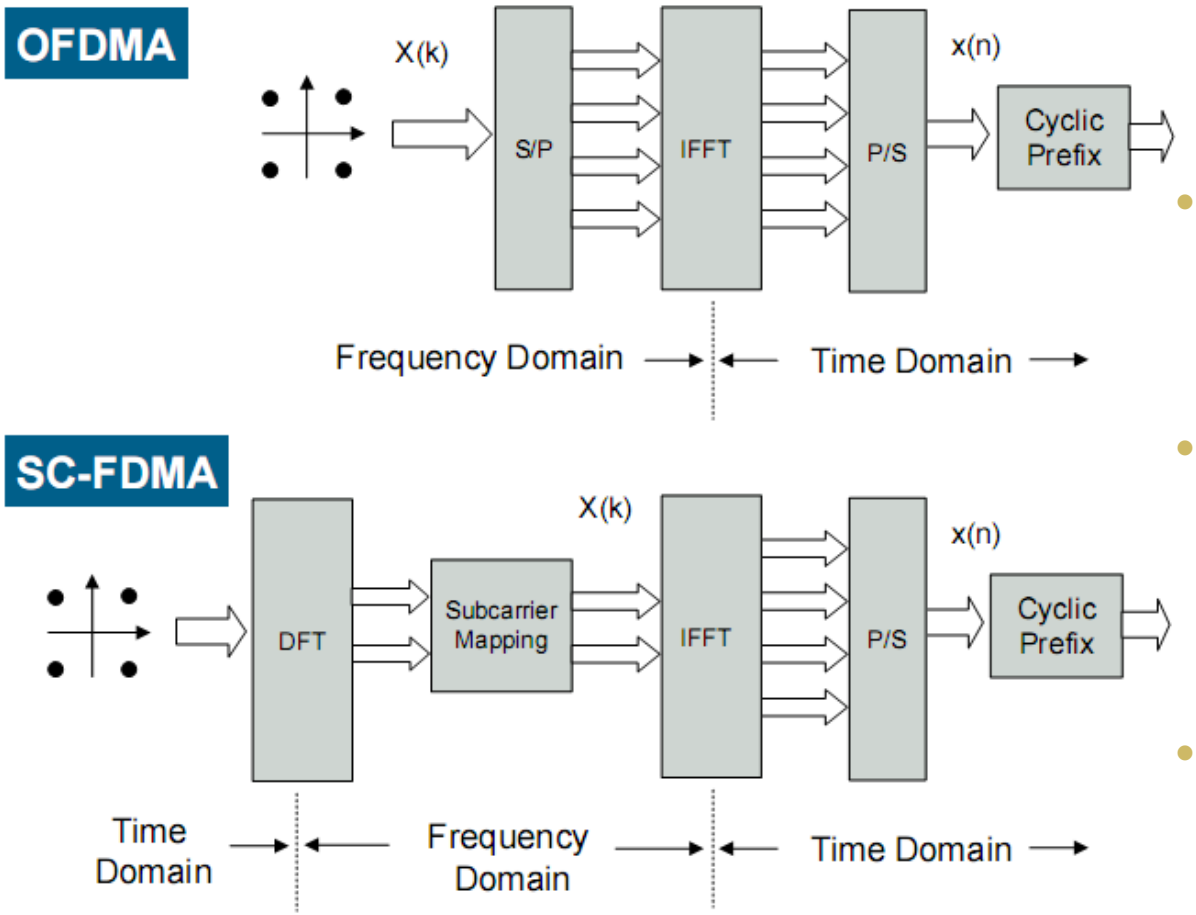
SC-FDMA

LTE Uplink Transmission Scheme: SC-FDMA

- Pemilihan OFDMA dianggap optimum untuk memenuhi persyaratan LTE pada arah downlink, tetapi OFDMA memiliki properti yang kurang menguntungkan pada arah Uplink.
- Hal tsb terutama disebabkan oleh lemahnya peak-to-average power ratio (PAPR) dari sinyal OFDMA, yang mengakibatkan buruknya coverage uplink.
- Oleh karena itu, skema transmisi Uplink LTE untuk mode FDD maupun TDD didasarkan pada SC-FDMA, yang mempunyai properti PAPR lebih baik.
- Pemrosesan sinyal SC-FDMA memiliki beberapa kesamaan dengan pemrosesan sinyal OFDMA, sehingga parameter-parameter DL dan UL dapat diharmonisasi.
- Untuk membangkitkan sinyal SC-FDMA, E-UTRA telah memilih DFT-spread-OFDM (DFT-s-OFDM).

Floatway Learning Center

OFDMA and SC-FDMA

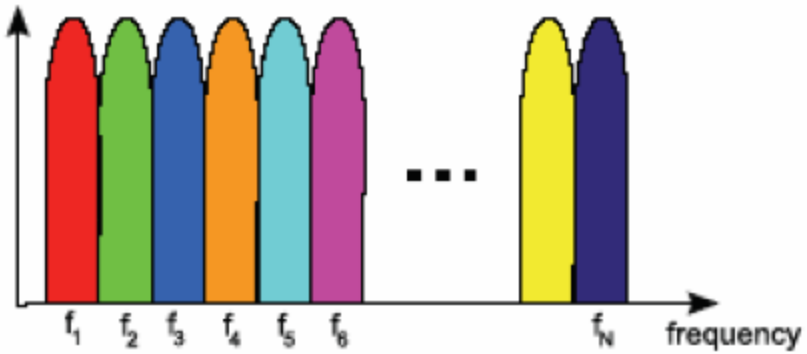


- The symbol mapping in OFDM happens in the frequency domain.
- In SC-FDMA, the symbol mapping is done in the time domain.
- Appropriate subscriber mapping in the frequency domain allows to control the PAPR.
- SC-FDMA enable frequency domain equalizer approaches like OFDMA

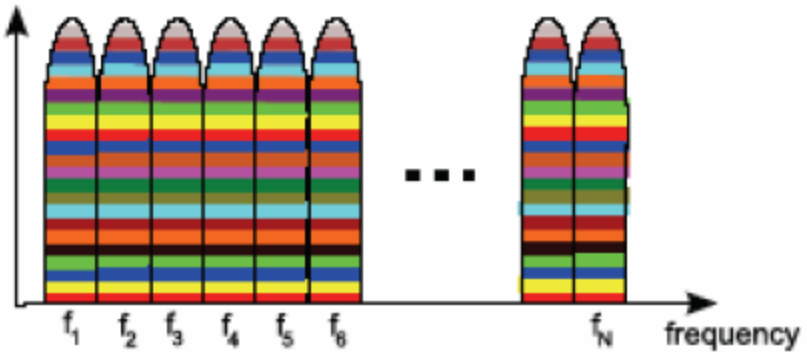
Floatway Learning Center

How does a SC-FDMA signal look like?

- Similar to OFDM signal, but…
  - …in OFDMA, each sub-carrier only carries information related to one specific symbol,
  - …in SC-FDMA, each sub-carrier contains information of ALL transmitted symbols.



(a) OFDM subcarriers



(b) DFT-s-OFDM subcarriers

Floatway Learning Center

SC-FDMA parameterization (FDD and TDD)

LTE FDD

•Same as in downlink

Configuration	Number SC-FDMA Symbols	Number of Subcarrier	Cyclic Prefix Length in Samples	Cyclic Prefix Length in µs
Normal CP $\Delta f = 15$ kHz	7	12	160 for 1st symbol 144 for other symbols	5.2 for 1st symbol 4.7 for other symbols
Extended CP $\Delta f = 15$ kHz	6		512	16.7

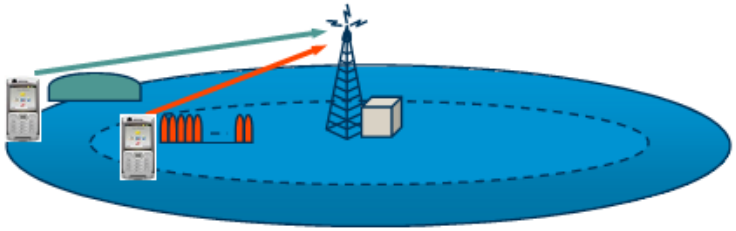
TD-LTE

•Usage of UL depends on the selected UL-DL configuration (1 to 8), each configuration offers a different number of subframes (1ms) for uplink transmission,

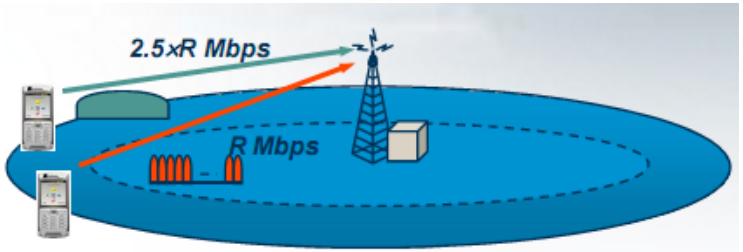
•Parameterization for those subframes, means number of SC-FDMA symbols same as for FDD and depending on CP,

Floatway Learning Center

Improved UL Performance
SC-FDMA compared to ordinary OFDM

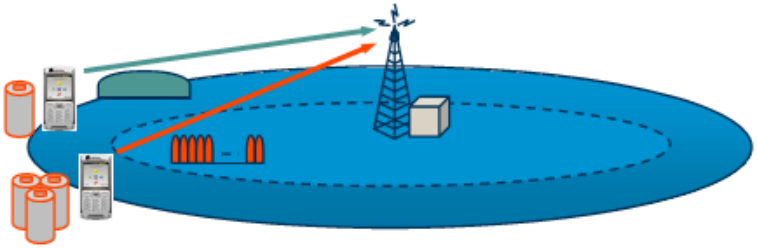


or



Improved coverage
(> 60% improvement)

Higher data rates
(> 2.5 times improvement)



Reduced power consumption
Longer battery life

Single-carrier transmission in uplink enables low PAPR that gives more 4 dB better link budget and reduced power consumption compared to OFDM

Floatway Learning Center

LTE Uplink SC-FDMA Physical Layer Parameters

Parameter	Value	Description
Subframe duration	1 ms	
Slot duration	0.5 ms	
Subcarrier spacing	15 kHz	
SC-FDMA symbol duration	66.67 μs	
CP duration	Normal CP:	5.2 μs first symbol in each slot, 4.69 μs all other symbols
	Extended CP:	16.67 μs all symbols
Number of symbols per slot	7 (Normal CP) 6 (Extended CP)	
Number of subcarriers per RB	12	

	Carrier bandwidth (MHz)					
	1.4	3	5	10	15	20
FFT size	128	256	512	1024	1536	2048
Sampling rate: M/N × 3.84 MHz	1/2	1/1	2/1	4/1	6/1	8/1
Number of subcarriers	72	180	300	600	900	1200
Number of RBs	6	15	25	50	75	100
Bandwidth efficiency (%)	77.1	90	90	90	90	90

Floatway Learning Center

Floatway Learning Center

MIMO ANTENNA

Different gains of multiple antenna systems

- ”Smart antenna” gain
  - Beamforming to increase the average signal-to-noise (SNR) ratio through focussing energy into desired directions
- Spatial diversity gain
  - Receiving on multiple antenna elements reduces fading problems. The diversity order is defined by the number of *decorrelated* spatial branches
- Spatial multiplexing gain
  - A *matrix* channel is created, opening up the possibility of transmitting over several spatial modes of the matrix channel increasing the link throughput at no additional frequency, timer or power expenditure

Floatway Learning Center

Data transmission over MIMO systems

- Two main categories:
  - Data rate maximization
    - Sending as many independent signals as antennas
    - Spatial multiplexing
  - Diversity maximization
    - The individual streams can be encoded jointly
    - Protect against transmission errors caused by channel fading
    - Minimize the outage probability

Floatway Learning Center

Quiz 5 : LTE DL Peak Rate

Hitunglah maksimum Peak Rate dari masing-masing Konfigurasi Antenna berikut (Modulasi 64 QAM).

Bandwidth (MHz)	Antenna		
	SISO	MIMO 2 x 2	MIMO 4 x 4
1.4			
3			
5			
10			
15			
20			

Floatway Learning Center

Floatway Learning Center

End of Training Session

FLOATWAY SYSTEMS